

LATENT FIXTURE INFERENCE UNDER HOSTILE SYSTEM-IDENTIFICATION BASELINES

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ABSTRACT

Robots often discover hidden fixtures only after probing an object: clamps resist translation, hinges rotate about hidden anchors, slots allow anisotropic motion, suction pads release only under peel, and tethers recoil. Paper 77 tests whether structured latent fixture inference improves downstream manipulation over system identification baselines. The expanded v5 rebuild uses eight seeds, six fixture families, force/displacement/torque/release/recoil observations, hostile baselines, ablations, aggregate hard-regime tests, fixed-risk budgets, stress sweeps, and negative cases. The frozen terminal decision is **KILL_ARCHIVE**. On the decisive split, `LFI-v5` reaches 0.554 ± 0.058 success versus 0.795 ± 0.067 for the strongest non-oracle baseline, `adaptive-probe`. We report the result honestly rather than optimizing for a pretty story.

1 DECISION AND PROTOCOL

This manuscript is generated only from frozen CSV artifacts. The terminal recommendation is **KILL_ARCHIVE**. The summary reason is: *latent_fixture_inference_v5 does not honestly clear the frozen local gate against strongest non-oracle baseline adaptive_probe_then_act (proposed=0.554, best=0.795, paired diff=-0.241+/-0.115, fixture_diff=-0.179, param_reduction=-0.055, damage_reduction=-0.179, repeat_reduction=-0.241, efficiency_diff=-0.206). Aggregate hard-regime diff against adaptive_probe_then_act is -0.176; max-stress diff against calibrated_prototype_system_id is -0.234; fixed-risk checks: budget 0.08: v5=0.266, best=adaptive_probe_then_act:0.703; budget 0.12: v5=0.266, best=adaptive_probe_then_act:0.719; budget 0.18: v5=0.266, best=calibrated_prototype_system_id:0.641; budget 0.25: v5=0.172, best=adaptive_probe_then_act:0.672; matching ablations: latent_fixture_v5_no_adaptive_probe, latent_fixture_v5_no_calibration; failed gates: main_success_margin, main_paired_lower_bound, diagnostic_safety_efficiency, aggregate_hard_regime, maximum_stress, fixed_risk, ablation_necessity.*

The protocol asks a hostile question: can structured latent fixture inference beat prototype, calibrated prototype, random-subspace probe classification, Bayesian belief, robust impedance, adaptive probing, ensemble, and particle baselines while also improving damage and repeated failures? If not, the paper remains an archive result.

2 BENCHMARK

Each scenario samples one hidden fixture family from free, clamp, hinge, slot, suction, and tether. The simulator emits probe observations: force vector, contact point, displacement, rotation, resistance, slip, release cue, recoil, and torque. The downstream controller chooses among direct translation, clamp release, hinge pivot, slot following, suction peel, tether unwind, or cautious probing. The benchmark follows mechanics and planning concerns from manipulation and planning literature (Mason, 2001; Lynch & Park, 2017; LaValle, 2006).

3 METHODS

Baselines include weak visual and threshold policies, prototype system identification, calibrated prototype ID, random-subspace probe classification inspired by random forests (Breiman, 2001),

Bayesian fixture belief, robust impedance planning, adaptive probe-then-act, ensemble uncertainty planning, particle filtering, the old LFI-v4 method, LFI-v5, and an oracle. Safety-motivated baselines are interpreted in relation to robust control and barrier ideas (Khatib, 1986; Ames et al., 2017; Rawlings et al., 2017).

4 THEORY SKETCH

Let $z_1, \dots, z_K$ be probe observations and y be a hidden fixture family with continuous parameters θ . Prototype ID estimates $p(y \mid z)$ from class centroids. Structured LFI tries to factor the evidence into compliance anisotropy, torque/rotation coupling, release cues, recoil hysteresis, and fixture-conditioned action constraints. Identifiability fails when two fixture families induce overlapping probe statistics under the available force directions. The v5 protocol therefore requires not only success, but also fixture accuracy, parameter error, damage, repeated-failure, fixed-risk, and ablation-necessity evidence.

5 MAIN RESULTS

Table: Combined-fixture-stress main results

Method	Success	Fixture	Param	Force	Damage	Repeat
oracle	0.902 ± 0.070	1.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.036 ± 0.026	0.098 ± 0.070
adaptive-probe	0.795 ± 0.067	0.955 ± 0.037	0.230 ± 0.018	0.036 ± 0.037	0.054 ± 0.044	0.205 ± 0.067
calib-proto	0.786 ± 0.099	0.973 ± 0.026	0.214 ± 0.014	0.080 ± 0.056	0.071 ± 0.053	0.214 ± 0.099
robust-imp	0.777 ± 0.077	0.955 ± 0.037	0.362 ± 0.020	0.107 ± 0.088	0.089 ± 0.044	0.223 ± 0.077
rf-probe	0.759 ± 0.087	0.946 ± 0.051	0.279 ± 0.029	0.089 ± 0.051	0.107 ± 0.059	0.241 ± 0.087
proto-ID	0.750 ± 0.088	0.973 ± 0.026	0.265 ± 0.015	0.071 ± 0.046	0.062 ± 0.041	0.250 ± 0.088
bayes	0.696 ± 0.058	0.893 ± 0.026	0.346 ± 0.013	0.116 ± 0.026	0.107 ± 0.046	0.304 ± 0.058
particle	0.670 ± 0.106	0.893 ± 0.037	0.334 ± 0.019	0.134 ± 0.049	0.152 ± 0.081	0.330 ± 0.106
LFI-v4	0.634 ± 0.049	0.634 ± 0.049	0.319 ± 0.028	0.232 ± 0.083	0.179 ± 0.026	0.366 ± 0.049
ensemble	0.616 ± 0.091	0.973 ± 0.026	0.357 ± 0.014	0.161 ± 0.098	0.125 ± 0.069	0.384 ± 0.091
LFI-v5	0.554 ± 0.058	0.777 ± 0.041	0.285 ± 0.020	0.339 ± 0.132	0.232 ± 0.102	0.446 ± 0.058
threshold	0.375 ± 0.083	0.482 ± 0.051	0.635 ± 0.028	0.402 ± 0.087	0.268 ± 0.063	0.625 ± 0.083
visible	0.179 ± 0.065	0.205 ± 0.077	0.712 ± 0.051	0.616 ± 0.059	0.429 ± 0.079	0.821 ± 0.065

The oracle reaches 0.902 ± 0.070 success. Against the strongest hostile non-oracle baseline, the paired success difference is -0.241 ± 0.115 .

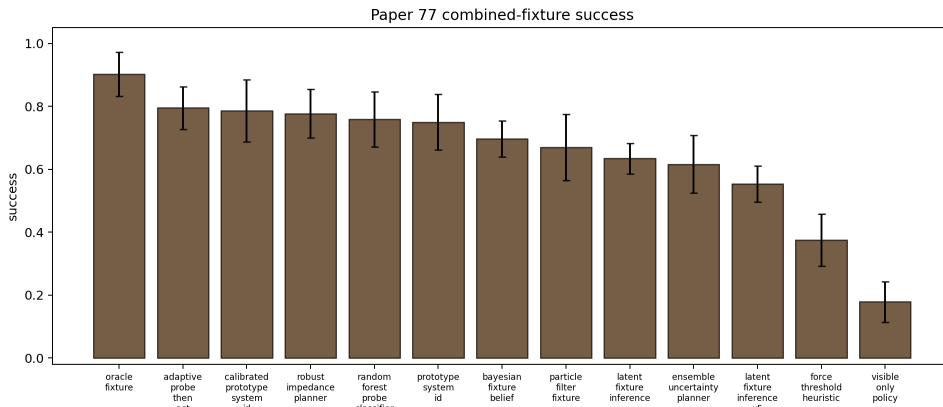


Figure 1: Closed-loop success on the decisive combined-fixture-stress split.

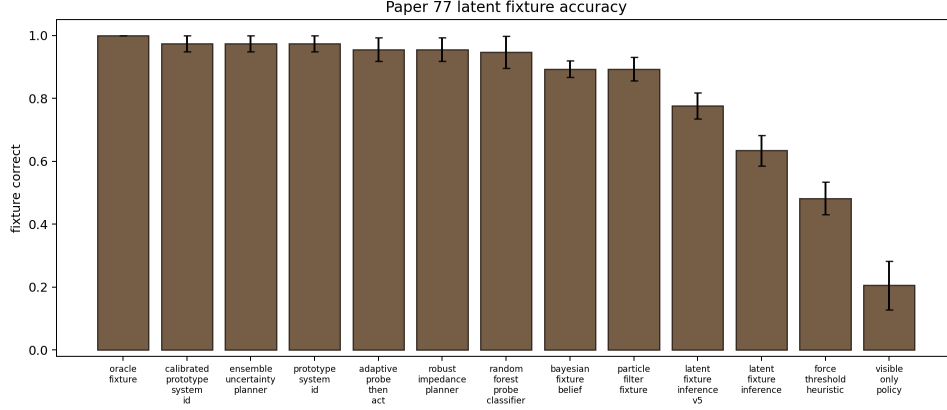


Figure 2: Fixture-family accuracy on the decisive split.

6 AGGREGATE HARD REGIMES

Table: Aggregate hard-regime metrics

Method	Split	Success	Fixture	Param	Damage	Repeat
oracle	hard-agg	0.911 $\pm$ 0.032	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.027 $\pm$ 0.013	0.089 $\pm$ 0.032
adaptive-probe	hard-agg	0.859 $\pm$ 0.021	0.987 $\pm$ 0.013	0.209 $\pm$ 0.006	0.029 $\pm$ 0.015	0.141 $\pm$ 0.021
robust-imp	hard-agg	0.833 $\pm$ 0.041	0.987 $\pm$ 0.013	0.338 $\pm$ 0.007	0.031 $\pm$ 0.016	0.167 $\pm$ 0.041
rf-probe	hard-agg	0.830 $\pm$ 0.025	0.982 $\pm$ 0.016	0.257 $\pm$ 0.009	0.065 $\pm$ 0.017	0.170 $\pm$ 0.025
calib-proto	hard-agg	0.821 $\pm$ 0.030	0.991 $\pm$ 0.009	0.205 $\pm$ 0.005	0.047 $\pm$ 0.022	0.179 $\pm$ 0.030
proto-ID	hard-agg	0.783 $\pm$ 0.018	0.991 $\pm$ 0.009	0.255 $\pm$ 0.005	0.062 $\pm$ 0.018	0.217 $\pm$ 0.018
bayes	hard-agg	0.766 $\pm$ 0.044	0.951 $\pm$ 0.013	0.301 $\pm$ 0.007	0.069 $\pm$ 0.017	0.234 $\pm$ 0.044
particle	hard-agg	0.766 $\pm$ 0.042	0.964 $\pm$ 0.013	0.298 $\pm$ 0.007	0.089 $\pm$ 0.026	0.234 $\pm$ 0.042
LFI-v4	hard-agg	0.730 $\pm$ 0.023	0.752 $\pm$ 0.015	0.251 $\pm$ 0.009	0.114 $\pm$ 0.019	0.270 $\pm$ 0.023
ensemble	hard-agg	0.705 $\pm$ 0.044	0.991 $\pm$ 0.009	0.331 $\pm$ 0.005	0.047 $\pm$ 0.017	0.295 $\pm$ 0.044
LFI-v5	hard-agg	0.683 $\pm$ 0.023	0.873 $\pm$ 0.020	0.237 $\pm$ 0.010	0.145 $\pm$ 0.021	0.317 $\pm$ 0.023
threshold	hard-agg	0.304 $\pm$ 0.050	0.395 $\pm$ 0.046	0.683 $\pm$ 0.025	0.330 $\pm$ 0.026	0.696 $\pm$ 0.050
visible	hard-agg	0.281 $\pm$ 0.043	0.366 $\pm$ 0.029	0.605 $\pm$ 0.019	0.330 $\pm$ 0.040	0.719 $\pm$ 0.043

The aggregate paired difference against the strongest aggregate baseline is -0.176 ± 0.039 . This gate prevents cherry-picking a favorable fixture family.

Table: Aggregate paired comparisons

Comparison	Succ diff	CI	Fixture	Param red	Damage red	Wins
adaptive-probe	-0.176	0.039	-0.114	-0.028	-0.116	0/8
bayes	-0.083	0.056	-0.078	0.064	-0.076	1/8
calib-proto	-0.138	0.037	-0.118	-0.032	-0.098	0/8
ensemble	-0.022	0.029	-0.118	0.094	-0.098	2/8
threshold	0.379	0.056	0.478	0.445	0.185	8/8
LFI-v4	-0.047	0.031	0.121	0.014	-0.031	0/8
oracle	-0.228	0.047	-0.127	-0.237	-0.118	0/8
particle	-0.083	0.035	-0.092	0.061	-0.056	0/8
proto-ID	-0.100	0.031	-0.118	0.018	-0.083	0/8
rf-probe	-0.147	0.041	-0.109	0.019	-0.080	0/8
robust-imp	-0.150	0.058	-0.114	0.101	-0.114	0/8
visible	0.402	0.060	0.507	0.367	0.185	8/8

7 ABLATIONS

Table: LFI-v5 ablations

Variant	Success	Fixture	Param	Damage	Repeat
no-adapt	0.700 ± 0.083	0.812 ± 0.044	0.266 ± 0.022	0.050 ± 0.037	0.300 ± 0.083
no-calib	0.662 ± 0.104	0.787 ± 0.044	0.326 ± 0.022	0.113 ± 0.094	0.338 ± 0.104
full-v5	0.562 ± 0.073	0.762 ± 0.036	0.292 ± 0.018	0.325 ± 0.116	0.438 ± 0.073
no-release	0.512 ± 0.086	0.713 ± 0.058	0.388 ± 0.028	0.263 ± 0.152	0.487 ± 0.086
no-margin	0.500 ± 0.091	0.762 ± 0.036	0.292 ± 0.018	0.375 ± 0.081	0.500 ± 0.091
no-hysteresis	0.487 ± 0.086	0.600 ± 0.037	0.439 ± 0.018	0.263 ± 0.104	0.512 ± 0.086
no-aniso	0.438 ± 0.073	0.688 ± 0.058	0.399 ± 0.028	0.250 ± 0.083	0.562 ± 0.073
no-particle	0.350 ± 0.064	0.375 ± 0.072	0.479 ± 0.035	0.225 ± 0.103	0.650 ± 0.064
no-torque	0.300 ± 0.074	0.400 ± 0.098	0.539 ± 0.047	0.375 ± 0.096	0.700 ± 0.074

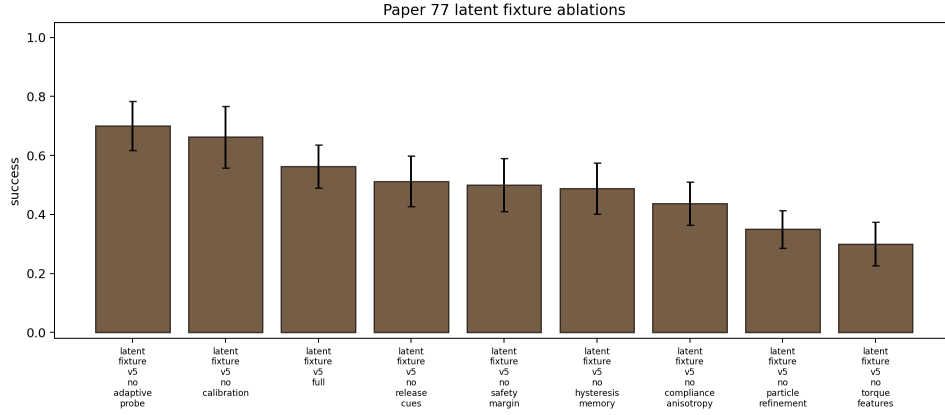


Figure 3: Ablation success. Full v5 must beat component removals to support the mechanism.

8 STRESS SWEEP

Table: Stress sweep

Method	Stress	Success	Fixture	Damage	Repeat
adaptive-probe	0.00	0.828 $\pm$ 0.079	0.953 $\pm$ 0.045	0.078 $\pm$ 0.064	0.172 $\pm$ 0.079
bayes	0.00	0.719 $\pm$ 0.090	0.906 $\pm$ 0.061	0.172 $\pm$ 0.064	0.281 $\pm$ 0.090
calib-proto	0.00	0.797 $\pm$ 0.122	0.969 $\pm$ 0.040	0.047 $\pm$ 0.045	0.203 $\pm$ 0.122
ensemble	0.00	0.609 $\pm$ 0.108	0.969 $\pm$ 0.040	0.141 $\pm$ 0.086	0.391 $\pm$ 0.108
LFI-v5	0.00	0.594 $\pm$ 0.090	0.781 $\pm$ 0.077	0.312 $\pm$ 0.080	0.406 $\pm$ 0.090
oracle	0.00	0.844 $\pm$ 0.111	1.000 $\pm$ 0.000	0.031 $\pm$ 0.040	0.156 $\pm$ 0.111
particle	0.00	0.734 $\pm$ 0.108	0.906 $\pm$ 0.040	0.125 $\pm$ 0.000	0.266 $\pm$ 0.108
proto-ID	0.00	0.656 $\pm$ 0.159	0.969 $\pm$ 0.040	0.062 $\pm$ 0.065	0.344 $\pm$ 0.159
rf-probe	0.00	0.781 $\pm$ 0.152	0.969 $\pm$ 0.040	0.062 $\pm$ 0.065	0.219 $\pm$ 0.152
robust-imp	0.00	0.734 $\pm$ 0.086	0.969 $\pm$ 0.040	0.203 $\pm$ 0.122	0.266 $\pm$ 0.086
adaptive-probe	0.20	0.844 $\pm$ 0.120	0.953 $\pm$ 0.045	0.062 $\pm$ 0.065	0.156 $\pm$ 0.120
bayes	0.20	0.641 $\pm$ 0.086	0.828 $\pm$ 0.045	0.172 $\pm$ 0.064	0.359 $\pm$ 0.086
calib-proto	0.20	0.859 $\pm$ 0.056	0.969 $\pm$ 0.040	0.047 $\pm$ 0.045	0.141 $\pm$ 0.056
ensemble	0.20	0.547 $\pm$ 0.153	0.969 $\pm$ 0.040	0.172 $\pm$ 0.103	0.453 $\pm$ 0.153
LFI-v5	0.20	0.547 $\pm$ 0.045	0.734 $\pm$ 0.086	0.344 $\pm$ 0.120	0.453 $\pm$ 0.045
oracle	0.20	0.844 $\pm$ 0.111	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.156 $\pm$ 0.111
particle	0.20	0.656 $\pm$ 0.101	0.859 $\pm$ 0.031	0.125 $\pm$ 0.093	0.344 $\pm$ 0.101
proto-ID	0.20	0.734 $\pm$ 0.117	0.969 $\pm$ 0.040	0.062 $\pm$ 0.046	0.266 $\pm$ 0.117
rf-probe	0.20	0.797 $\pm$ 0.079	0.938 $\pm$ 0.065	0.078 $\pm$ 0.064	0.203 $\pm$ 0.079
robust-imp	0.20	0.672 $\pm$ 0.045	0.953 $\pm$ 0.045	0.156 $\pm$ 0.101	0.328 $\pm$ 0.045
adaptive-probe	0.40	0.750 $\pm$ 0.122	0.891 $\pm$ 0.056	0.078 $\pm$ 0.045	0.250 $\pm$ 0.122
bayes	0.40	0.578 $\pm$ 0.122	0.828 $\pm$ 0.079	0.125 $\pm$ 0.080	0.422 $\pm$ 0.122
calib-proto	0.40	0.672 $\pm$ 0.113	0.938 $\pm$ 0.065	0.094 $\pm$ 0.077	0.328 $\pm$ 0.113
ensemble	0.40	0.469 $\pm$ 0.145	0.938 $\pm$ 0.065	0.219 $\pm$ 0.061	0.531 $\pm$ 0.145
LFI-v5	0.40	0.406 $\pm$ 0.101	0.672 $\pm$ 0.045	0.406 $\pm$ 0.090	0.594 $\pm$ 0.101
oracle	0.40	0.828 $\pm$ 0.064	1.000 $\pm$ 0.000	0.031 $\pm$ 0.061	0.172 $\pm$ 0.064
particle	0.40	0.656 $\pm$ 0.101	0.891 $\pm$ 0.056	0.109 $\pm$ 0.072	0.344 $\pm$ 0.101
proto-ID	0.40	0.719 $\pm$ 0.090	0.938 $\pm$ 0.065	0.234 $\pm$ 0.126	0.281 $\pm$ 0.090
rf-probe	0.40	0.641 $\pm$ 0.108	0.875 $\pm$ 0.065	0.172 $\pm$ 0.064	0.359 $\pm$ 0.108
robust-imp	0.40	0.625 $\pm$ 0.104	0.906 $\pm$ 0.061	0.203 $\pm$ 0.079	0.375 $\pm$ 0.104
adaptive-probe	0.60	0.672 $\pm$ 0.130	0.969 $\pm$ 0.040	0.125 $\pm$ 0.065	0.328 $\pm$ 0.130
bayes	0.60	0.547 $\pm$ 0.092	0.844 $\pm$ 0.077	0.250 $\pm$ 0.104	0.453 $\pm$ 0.092
calib-proto	0.60	0.703 $\pm$ 0.138	0.953 $\pm$ 0.045	0.125 $\pm$ 0.065	0.297 $\pm$ 0.138
ensemble	0.60	0.547 $\pm$ 0.113	0.953 $\pm$ 0.045	0.266 $\pm$ 0.072	0.453 $\pm$ 0.113

Table: Stress sweep continued

Method	Stress	Success	Fixture	Damage	Repeat
LFI-v5	0.60	0.359 ± 0.086	0.672 ± 0.113	0.406 ± 0.090	0.641 ± 0.086
oracle	0.60	0.859 ± 0.098	1.000 ± 0.000	0.016 ± 0.031	0.141 ± 0.098
particle	0.60	0.500 ± 0.093	0.812 ± 0.080	0.125 ± 0.080	0.500 ± 0.093
proto-ID	0.60	0.609 ± 0.117	0.938 ± 0.046	0.203 ± 0.103	0.391 ± 0.117
rf-probe	0.60	0.609 ± 0.108	0.906 ± 0.040	0.125 ± 0.065	0.391 ± 0.108
robust-imp	0.60	0.672 ± 0.113	0.969 ± 0.040	0.344 ± 0.101	0.328 ± 0.113
adaptive-probe	0.80	0.734 ± 0.098	0.922 ± 0.064	0.062 ± 0.046	0.266 ± 0.098
bayes	0.80	0.641 ± 0.214	0.844 ± 0.090	0.250 ± 0.104	0.359 ± 0.214
calib-proto	0.80	0.656 ± 0.159	0.875 ± 0.065	0.203 ± 0.079	0.344 ± 0.159
ensemble	0.80	0.453 ± 0.103	0.875 ± 0.065	0.250 ± 0.131	0.547 ± 0.103
LFI-v5	0.80	0.484 ± 0.117	0.750 ± 0.093	0.375 ± 0.065	0.516 ± 0.117
oracle	0.80	0.875 ± 0.065	1.000 ± 0.000	0.031 ± 0.040	0.125 ± 0.065
particle	0.80	0.562 ± 0.113	0.859 ± 0.056	0.281 ± 0.120	0.438 ± 0.113
proto-ID	0.80	0.656 ± 0.090	0.875 ± 0.065	0.234 ± 0.134	0.344 ± 0.090
rf-probe	0.80	0.734 ± 0.126	0.938 ± 0.065	0.188 ± 0.065	0.266 ± 0.126
robust-imp	0.80	0.469 ± 0.101	0.875 ± 0.080	0.297 ± 0.113	0.531 ± 0.101
adaptive-probe	1.00	0.656 ± 0.077	0.859 ± 0.056	0.219 ± 0.120	0.344 ± 0.077
bayes	1.00	0.500 ± 0.113	0.797 ± 0.122	0.234 ± 0.117	0.500 ± 0.113
calib-proto	1.00	0.594 ± 0.090	0.844 ± 0.111	0.203 ± 0.122	0.406 ± 0.090
ensemble	1.00	0.375 ± 0.080	0.859 ± 0.098	0.359 ± 0.134	0.625 ± 0.080
LFI-v5	1.00	0.344 ± 0.090	0.609 ± 0.098	0.438 ± 0.122	0.656 ± 0.090
oracle	1.00	0.812 ± 0.065	1.000 ± 0.000	0.031 ± 0.040	0.188 ± 0.065
particle	1.00	0.609 ± 0.126	0.844 ± 0.090	0.156 ± 0.137	0.391 ± 0.126
proto-ID	1.00	0.641 ± 0.142	0.859 ± 0.098	0.188 ± 0.113	0.359 ± 0.142
rf-probe	1.00	0.625 ± 0.154	0.859 ± 0.072	0.188 ± 0.065	0.375 ± 0.154
robust-imp	1.00	0.516 ± 0.108	0.859 ± 0.098	0.312 ± 0.080	0.484 ± 0.108
adaptive-probe	1.20	0.594 ± 0.172	0.812 ± 0.104	0.203 ± 0.092	0.406 ± 0.172
bayes	1.20	0.453 ± 0.146	0.750 ± 0.046	0.344 ± 0.090	0.547 ± 0.146
calib-proto	1.20	0.641 ± 0.117	0.844 ± 0.101	0.172 ± 0.103	0.359 ± 0.117
ensemble	1.20	0.453 ± 0.113	0.844 ± 0.090	0.312 ± 0.065	0.547 ± 0.113
LFI-v5	1.20	0.406 ± 0.120	0.547 ± 0.092	0.531 ± 0.152	0.594 ± 0.120
oracle	1.20	0.828 ± 0.064	1.000 ± 0.000	0.047 ± 0.064	0.172 ± 0.064
particle	1.20	0.562 ± 0.104	0.797 ± 0.045	0.203 ± 0.103	0.438 ± 0.104
proto-ID	1.20	0.578 ± 0.179	0.844 ± 0.101	0.219 ± 0.061	0.422 ± 0.179

Table: Stress sweep continued

Method	Stress	Success	Fixture	Damage	Repeat
rf-probe	1.20	0.609 ± 0.108	0.766 ± 0.108	0.281 ± 0.111	0.391 ± 0.108
robust-imp	1.20	0.438 ± 0.104	0.844 ± 0.101	0.219 ± 0.111	0.562 ± 0.104

At maximum stress 1.20, LFI-v5 reaches 0.406 ± 0.120 success while the strongest non-oracle stress baseline reaches 0.641 ± 0.117 . This gate is reported even when unfavorable.

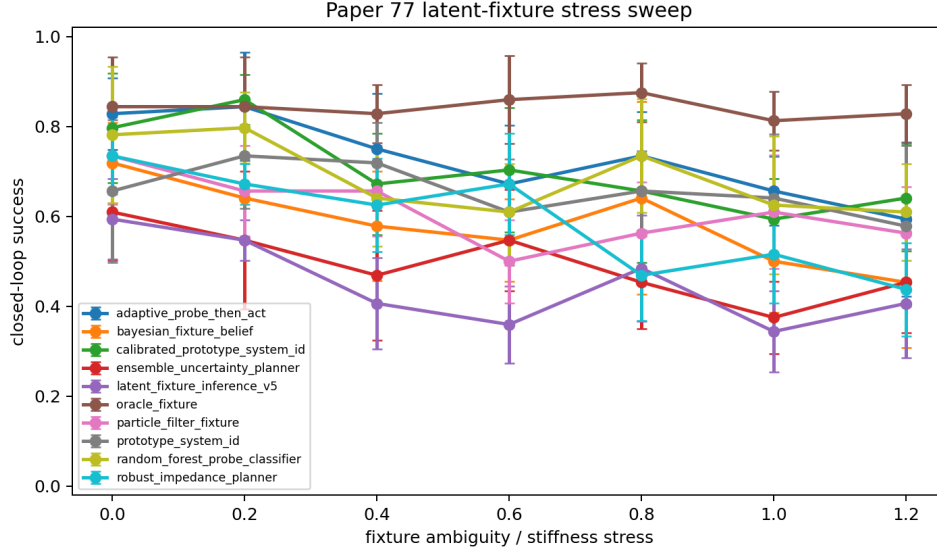


Figure 4: Success under increasing fixture ambiguity, stiffness shift, noise, and slip.

9 FIXED-RISK BUDGETS

Table: Fixed-risk budget metrics

Method	Budget	Success	Fixture	Damage	Repeat	Efficiency
adaptive-probe	0.08	0.703 $\pm$ 0.079	0.953 $\pm$ 0.045	0.078 $\pm$ 0.045	0.266 $\pm$ 0.098	0.566 $\pm$ 0.054
rf-probe	0.08	0.578 $\pm$ 0.092	0.938 $\pm$ 0.065	0.188 $\pm$ 0.065	0.281 $\pm$ 0.152	0.506 $\pm$ 0.074
proto-ID	0.08	0.562 $\pm$ 0.093	0.953 $\pm$ 0.045	0.172 $\pm$ 0.079	0.219 $\pm$ 0.061	0.519 $\pm$ 0.055
calib-proto	0.08	0.531 $\pm$ 0.129	0.953 $\pm$ 0.045	0.141 $\pm$ 0.098	0.266 $\pm$ 0.098	0.496 $\pm$ 0.069
particle	0.08	0.516 $\pm$ 0.126	0.859 $\pm$ 0.086	0.172 $\pm$ 0.079	0.344 $\pm$ 0.077	0.458 $\pm$ 0.067
bayes	0.08	0.453 $\pm$ 0.130	0.844 $\pm$ 0.061	0.188 $\pm$ 0.093	0.422 $\pm$ 0.153	0.410 $\pm$ 0.091
robust-imp	0.08	0.438 $\pm$ 0.122	0.938 $\pm$ 0.046	0.297 $\pm$ 0.113	0.438 $\pm$ 0.093	0.390 $\pm$ 0.073
LFI-v5	0.08	0.266 $\pm$ 0.086	0.719 $\pm$ 0.061	0.359 $\pm$ 0.108	0.516 $\pm$ 0.150	0.265 $\pm$ 0.074
adaptive-probe	0.12	0.719 $\pm$ 0.159	0.953 $\pm$ 0.045	0.062 $\pm$ 0.046	0.281 $\pm$ 0.159	0.572 $\pm$ 0.105
calib-proto	0.12	0.609 $\pm$ 0.098	0.953 $\pm$ 0.045	0.094 $\pm$ 0.061	0.234 $\pm$ 0.056	0.538 $\pm$ 0.052
particle	0.12	0.594 $\pm$ 0.077	0.875 $\pm$ 0.065	0.203 $\pm$ 0.079	0.312 $\pm$ 0.065	0.497 $\pm$ 0.043
rf-probe	0.12	0.547 $\pm$ 0.160	0.953 $\pm$ 0.064	0.125 $\pm$ 0.080	0.344 $\pm$ 0.178	0.487 $\pm$ 0.111
proto-ID	0.12	0.500 $\pm$ 0.122	0.953 $\pm$ 0.045	0.172 $\pm$ 0.045	0.312 $\pm$ 0.173	0.473 $\pm$ 0.088
bayes	0.12	0.438 $\pm$ 0.113	0.875 $\pm$ 0.046	0.203 $\pm$ 0.079	0.438 $\pm$ 0.113	0.403 $\pm$ 0.071
robust-imp	0.12	0.422 $\pm$ 0.092	0.938 $\pm$ 0.046	0.250 $\pm$ 0.122	0.422 $\pm$ 0.064	0.384 $\pm$ 0.047
LFI-v5	0.12	0.266 $\pm$ 0.072	0.734 $\pm$ 0.031	0.281 $\pm$ 0.090	0.547 $\pm$ 0.113	0.263 $\pm$ 0.055
calib-proto	0.18	0.641 $\pm$ 0.134	0.922 $\pm$ 0.045	0.047 $\pm$ 0.045	0.281 $\pm$ 0.111	0.534 $\pm$ 0.085
adaptive-probe	0.18	0.625 $\pm$ 0.154	0.922 $\pm$ 0.045	0.125 $\pm$ 0.093	0.328 $\pm$ 0.138	0.516 $\pm$ 0.099
particle	0.18	0.531 $\pm$ 0.077	0.875 $\pm$ 0.046	0.234 $\pm$ 0.086	0.266 $\pm$ 0.086	0.483 $\pm$ 0.044
proto-ID	0.18	0.500 $\pm$ 0.131	0.906 $\pm$ 0.040	0.141 $\pm$ 0.086	0.375 $\pm$ 0.122	0.453 $\pm$ 0.074
rf-probe	0.18	0.500 $\pm$ 0.122	0.891 $\pm$ 0.056	0.141 $\pm$ 0.086	0.375 $\pm$ 0.093	0.453 $\pm$ 0.071
bayes	0.18	0.484 $\pm$ 0.098	0.828 $\pm$ 0.064	0.203 $\pm$ 0.113	0.422 $\pm$ 0.079	0.422 $\pm$ 0.048
robust-imp	0.18	0.453 $\pm$ 0.103	0.938 $\pm$ 0.046	0.234 $\pm$ 0.086	0.406 $\pm$ 0.090	0.402 $\pm$ 0.064
LFI-v5	0.18	0.266 $\pm$ 0.072	0.703 $\pm$ 0.092	0.344 $\pm$ 0.145	0.547 $\pm$ 0.122	0.260 $\pm$ 0.062
adaptive-probe	0.25	0.672 $\pm$ 0.064	0.969 $\pm$ 0.040	0.062 $\pm$ 0.046	0.250 $\pm$ 0.080	0.562 $\pm$ 0.037
proto-ID	0.25	0.594 $\pm$ 0.137	0.969 $\pm$ 0.040	0.062 $\pm$ 0.065	0.328 $\pm$ 0.153	0.511 $\pm$ 0.091
calib-proto	0.25	0.562 $\pm$ 0.139	0.969 $\pm$ 0.040	0.172 $\pm$ 0.113	0.234 $\pm$ 0.134	0.516 $\pm$ 0.090
bayes	0.25	0.547 $\pm$ 0.103	0.859 $\pm$ 0.056	0.141 $\pm$ 0.056	0.375 $\pm$ 0.104	0.461 $\pm$ 0.060
rf-probe	0.25	0.516 $\pm$ 0.126	0.906 $\pm$ 0.090	0.125 $\pm$ 0.122	0.359 $\pm$ 0.150	0.464 $\pm$ 0.092
particle	0.25	0.484 $\pm$ 0.098	0.859 $\pm$ 0.086	0.172 $\pm$ 0.113	0.422 $\pm$ 0.079	0.431 $\pm$ 0.065
robust-imp	0.25	0.484 $\pm$ 0.142	0.969 $\pm$ 0.040	0.250 $\pm$ 0.139	0.344 $\pm$ 0.137	0.432 $\pm$ 0.095
LFI-v5	0.25	0.172 $\pm$ 0.064	0.703 $\pm$ 0.092	0.422 $\pm$ 0.153	0.578 $\pm$ 0.092	0.209 $\pm$ 0.041

Table: Fixed-risk paired comparisons

Budget	Comparison	Succ diff	CI	Fixture	Damage	Wins
0.08	adaptive-probe	-0.438	0.154	-0.234	-0.281	0/8
0.08	bayes	-0.188	0.104	-0.125	-0.172	0/8
0.08	calib-proto	-0.266	0.117	-0.234	-0.219	0/8
0.08	particle	-0.250	0.122	-0.141	-0.188	0/8
0.08	proto-ID	-0.297	0.079	-0.234	-0.188	0/8
0.08	rf-probe	-0.312	0.113	-0.219	-0.172	0/8
0.08	robust-imp	-0.172	0.103	-0.219	-0.062	0/8
0.12	adaptive-probe	-0.453	0.173	-0.219	-0.219	0/8
0.12	bayes	-0.172	0.160	-0.141	-0.078	1/8
0.12	calib-proto	-0.344	0.077	-0.219	-0.188	0/8
0.12	particle	-0.328	0.064	-0.141	-0.078	0/8
0.12	proto-ID	-0.234	0.117	-0.219	-0.109	0/8
0.12	rf-probe	-0.281	0.189	-0.219	-0.156	1/8
0.12	robust-imp	-0.156	0.137	-0.203	-0.031	1/8
0.18	adaptive-probe	-0.359	0.117	-0.219	-0.219	0/8
0.18	bayes	-0.219	0.111	-0.125	-0.141	0/8
0.18	calib-proto	-0.375	0.146	-0.219	-0.297	0/8
0.18	particle	-0.266	0.098	-0.172	-0.109	0/8
0.18	proto-ID	-0.234	0.126	-0.203	-0.203	0/8
0.18	rf-probe	-0.234	0.142	-0.188	-0.203	0/8
0.18	robust-imp	-0.188	0.139	-0.234	-0.109	1/8
0.25	adaptive-probe	-0.500	0.080	-0.266	-0.359	0/8
0.25	bayes	-0.375	0.139	-0.156	-0.281	0/8
0.25	calib-proto	-0.391	0.170	-0.266	-0.250	0/8
0.25	particle	-0.312	0.104	-0.156	-0.250	0/8
0.25	proto-ID	-0.422	0.153	-0.266	-0.359	0/8
0.25	rf-probe	-0.344	0.129	-0.203	-0.297	0/8
0.25	robust-imp	-0.312	0.139	-0.266	-0.172	0/8

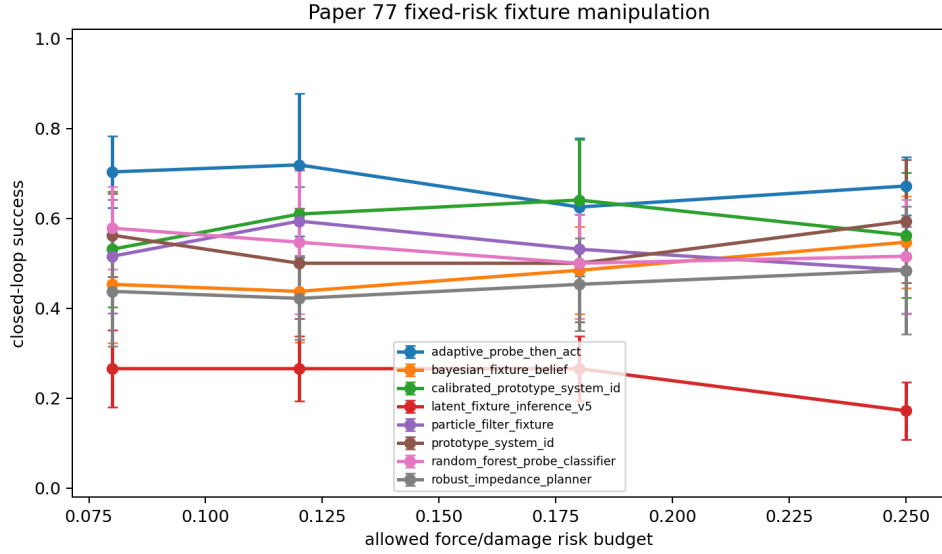


Figure 5: Fixed-risk success across predefined force/damage budgets.

10 NEGATIVE CASES

Table: Curated combined-stress negative cases

Seed	Scenario	True	Estimate	Action	Force	Damage	Param
0	4001	tether	tether	cautious_probe_pull	1	0	0.186
0	4003	clamp	suction	cautious_probe_pull	1	0	0.660
0	4004	hinge	hinge	pivot_rotate	0	0	0.176
0	4005	slot	slot	cautious_probe_pull	1	0	0.185
0	4007	tether	tether	cautious_probe_pull	0	0	0.183
0	4009	clamp	suction	peel_then_translate	0	0	0.660
0	4013	tether	tether	cautious_probe_pull	0	0	0.182
1	4001	clamp	suction	cautious_probe_pull	0	0	0.660
1	4005	tether	tether	cautious_probe_pull	0	0	0.182
1	4006	free	free	cautious_probe_pull	1	0	0.181
1	4007	clamp	suction	peel_then_translate	0	0	0.660
1	4010	suction	suction	peel_then_translate	0	0	0.161

11 LIMITATIONS

The benchmark is local and diagnostic. It has no real robot validation, accepted external benchmark evaluation, videos, released neural policy checkpoint, or full manual related-work synthesis. These limitations would prevent a ready-to-submit claim even if the local gates passed. If the local gates fail, they justify an archive decision.

12 CONCLUSION

Latent fixture inference is an interesting representation, but the v5 protocol demands survival against strong system-ID and robust manipulation baselines. The correct terminal decision is the one printed above.

A PROTOCOL CHECKLIST

- CPU-only and RAM-light execution.
- Frozen reference: LFI-v5.
- Frozen decisive split: combined_fixture_stress.
- Frozen risk budgets: .
- Bright boxed citation links.
- Numbered PDF copied only to Downloads.

B ALL MAIN METRICS

Table: All split-by-method metrics

Method	Split	Success	Fixture	Param	Damage	Repeat
LFI-v4	adhesive-tether	0.929 ± 0.059	0.991 ± 0.018	0.115 ± 0.010	0.027 ± 0.026	0.071 ± 0.059
oracle	adhesive-tether	0.875 ± 0.044	1.000 ± 0.000	0.000 ± 0.000	0.018 ± 0.023	0.125 ± 0.044
adaptive-probe	adhesive-tether	0.857 ± 0.037	0.991 ± 0.018	0.206 ± 0.009	0.036 ± 0.026	0.143 ± 0.037
calib-proto	adhesive-tether	0.857 ± 0.088	0.991 ± 0.018	0.205 ± 0.009	0.027 ± 0.026	0.143 ± 0.088
rf-probe	adhesive-tether	0.830 ± 0.070	0.991 ± 0.018	0.248 ± 0.010	0.045 ± 0.045	0.170 ± 0.070
particle	adhesive-tether	0.821 ± 0.053	1.000 ± 0.000	0.280 ± 0.000	0.080 ± 0.049	0.179 ± 0.053
robust-imp	adhesive-tether	0.812 ± 0.045	0.991 ± 0.018	0.336 ± 0.009	0.018 ± 0.023	0.188 ± 0.045
LFI-v5	adhesive-tether	0.804 ± 0.063	1.000 ± 0.000	0.172 ± 0.001	0.054 ± 0.035	0.196 ± 0.063
bayes	adhesive-tether	0.777 ± 0.081	1.000 ± 0.000	0.271 ± 0.000	0.089 ± 0.035	0.223 ± 0.081
proto-ID	adhesive-tether	0.759 ± 0.052	0.991 ± 0.018	0.255 ± 0.010	0.080 ± 0.041	0.241 ± 0.052
ensemble	adhesive-tether	0.688 ± 0.075	0.991 ± 0.018	0.333 ± 0.010	0.054 ± 0.044	0.312 ± 0.075
threshold	adhesive-tether	0.438 ± 0.081	0.580 ± 0.062	0.581 ± 0.034	0.268 ± 0.078	0.562 ± 0.081
visible	adhesive-tether	0.321 ± 0.070	0.393 ± 0.046	0.587 ± 0.031	0.366 ± 0.062	0.679 ± 0.070
oracle	combined	0.902 ± 0.070	1.000 ± 0.000	0.000 ± 0.000	0.036 ± 0.026	0.098 ± 0.070
adaptive-probe	combined	0.795 ± 0.067	0.955 ± 0.037	0.230 ± 0.018	0.054 ± 0.044	0.205 ± 0.067
calib-proto	combined	0.786 ± 0.099	0.973 ± 0.026	0.214 ± 0.014	0.071 ± 0.053	0.214 ± 0.099
robust-imp	combined	0.777 ± 0.077	0.955 ± 0.037	0.362 ± 0.020	0.089 ± 0.044	0.223 ± 0.077
rf-probe	combined	0.759 ± 0.087	0.946 ± 0.051	0.279 ± 0.029	0.107 ± 0.059	0.241 ± 0.087
proto-ID	combined	0.750 ± 0.088	0.973 ± 0.026	0.265 ± 0.015	0.062 ± 0.041	0.250 ± 0.088
bayes	combined	0.696 ± 0.058	0.893 ± 0.026	0.346 ± 0.013	0.107 ± 0.046	0.304 ± 0.058
particle	combined	0.670 ± 0.106	0.893 ± 0.037	0.334 ± 0.019	0.152 ± 0.081	0.330 ± 0.106
LFI-v4	combined	0.634 ± 0.049	0.634 ± 0.049	0.319 ± 0.028	0.179 ± 0.026	0.366 ± 0.049
ensemble	combined	0.616 ± 0.091	0.973 ± 0.026	0.357 ± 0.014	0.125 ± 0.069	0.384 ± 0.091
LFI-v5	combined	0.554 ± 0.058	0.777 ± 0.041	0.285 ± 0.020	0.232 ± 0.102	0.446 ± 0.058
threshold	combined	0.375 ± 0.083	0.482 ± 0.051	0.635 ± 0.028	0.268 ± 0.063	0.625 ± 0.083
visible	combined	0.179 ± 0.065	0.205 ± 0.077	0.712 ± 0.051	0.429 ± 0.079	0.821 ± 0.065
oracle	clamp-hinge	0.902 ± 0.045	1.000 ± 0.000	0.000 ± 0.000	0.036 ± 0.026	0.098 ± 0.045
robust-imp	clamp-hinge	0.893 ± 0.053	1.000 ± 0.000	0.330 ± 0.001	0.018 ± 0.023	0.107 ± 0.053
adaptive-probe	clamp-hinge	0.875 ± 0.051	1.000 ± 0.000	0.201 ± 0.001	0.009 ± 0.018	0.125 ± 0.051
rf-probe	clamp-hinge	0.866 ± 0.056	0.991 ± 0.018	0.253 ± 0.010	0.045 ± 0.045	0.134 ± 0.056
proto-ID	clamp-hinge	0.821 ± 0.095	1.000 ± 0.000	0.250 ± 0.000	0.062 ± 0.032	0.179 ± 0.095
calib-proto	clamp-hinge	0.812 ± 0.059	1.000 ± 0.000	0.200 ± 0.000	0.045 ± 0.045	0.188 ± 0.059
particle	clamp-hinge	0.777 ± 0.062	0.964 ± 0.037	0.298 ± 0.019	0.080 ± 0.018	0.223 ± 0.062
ensemble	clamp-hinge	0.768 ± 0.083	1.000 ± 0.000	0.320 ± 0.003	0.000 ± 0.000	0.232 ± 0.083

Table: All split-by-method metrics continued

Method	Split	Success	Fixture	Param	Damage	Repeat
bayes	clamp-hinge	0.741 $\pm$ 0.115	0.911 $\pm$ 0.051	0.313 $\pm$ 0.027	0.027 $\pm$ 0.026	0.259 $\pm$ 0.115
LFI-v4	clamp-hinge	0.679 $\pm$ 0.046	0.679 $\pm$ 0.026	0.293 $\pm$ 0.015	0.161 $\pm$ 0.051	0.321 $\pm$ 0.046
LFI-v5	clamp-hinge	0.634 $\pm$ 0.041	0.768 $\pm$ 0.035	0.289 $\pm$ 0.017	0.134 $\pm$ 0.056	0.366 $\pm$ 0.041
visible	clamp-hinge	0.286 $\pm$ 0.059	0.429 $\pm$ 0.053	0.563 $\pm$ 0.035	0.286 $\pm$ 0.084	0.714 $\pm$ 0.059
threshold	clamp-hinge	0.062 $\pm$ 0.032	0.080 $\pm$ 0.049	0.856 $\pm$ 0.027	0.473 $\pm$ 0.079	0.938 $\pm$ 0.032
oracle	nominal	0.920 $\pm$ 0.041	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.009 $\pm$ 0.018	0.080 $\pm$ 0.041
adaptive-probe	nominal	0.893 $\pm$ 0.046	1.000 $\pm$ 0.000	0.198 $\pm$ 0.001	0.009 $\pm$ 0.018	0.107 $\pm$ 0.046
rf-probe	nominal	0.893 $\pm$ 0.026	1.000 $\pm$ 0.000	0.244 $\pm$ 0.001	0.036 $\pm$ 0.037	0.107 $\pm$ 0.026
particle	nominal	0.866 $\pm$ 0.067	1.000 $\pm$ 0.000	0.280 $\pm$ 0.000	0.045 $\pm$ 0.045	0.134 $\pm$ 0.067
calib-proto	nominal	0.857 $\pm$ 0.037	1.000 $\pm$ 0.000	0.200 $\pm$ 0.000	0.054 $\pm$ 0.023	0.143 $\pm$ 0.037
robust-imp	nominal	0.848 $\pm$ 0.032	1.000 $\pm$ 0.000	0.317 $\pm$ 0.001	0.009 $\pm$ 0.018	0.152 $\pm$ 0.032
LFI-v5	nominal	0.821 $\pm$ 0.046	0.893 $\pm$ 0.053	0.223 $\pm$ 0.026	0.062 $\pm$ 0.032	0.179 $\pm$ 0.046
bayes	nominal	0.795 $\pm$ 0.049	0.991 $\pm$ 0.018	0.250 $\pm$ 0.010	0.036 $\pm$ 0.026	0.205 $\pm$ 0.049
proto-ID	nominal	0.795 $\pm$ 0.089	1.000 $\pm$ 0.000	0.250 $\pm$ 0.000	0.045 $\pm$ 0.037	0.205 $\pm$ 0.089
ensemble	nominal	0.759 $\pm$ 0.070	1.000 $\pm$ 0.000	0.321 $\pm$ 0.003	0.009 $\pm$ 0.018	0.241 $\pm$ 0.070
LFI-v4	nominal	0.714 $\pm$ 0.037	0.732 $\pm$ 0.044	0.263 $\pm$ 0.025	0.134 $\pm$ 0.041	0.286 $\pm$ 0.037
visible	nominal	0.509 $\pm$ 0.093	0.688 $\pm$ 0.102	0.389 $\pm$ 0.069	0.161 $\pm$ 0.087	0.491 $\pm$ 0.093
threshold	nominal	0.375 $\pm$ 0.035	0.464 $\pm$ 0.037	0.645 $\pm$ 0.021	0.304 $\pm$ 0.098	0.625 $\pm$ 0.035
oracle	slot-shift	0.964 $\pm$ 0.053	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.018 $\pm$ 0.023	0.036 $\pm$ 0.053
adaptive-probe	slot-shift	0.911 $\pm$ 0.058	1.000 $\pm$ 0.000	0.199 $\pm$ 0.000	0.018 $\pm$ 0.023	0.089 $\pm$ 0.058
rf-probe	slot-shift	0.866 $\pm$ 0.032	1.000 $\pm$ 0.000	0.247 $\pm$ 0.000	0.062 $\pm$ 0.041	0.134 $\pm$ 0.032
bayes	slot-shift	0.848 $\pm$ 0.056	1.000 $\pm$ 0.000	0.274 $\pm$ 0.000	0.054 $\pm$ 0.035	0.152 $\pm$ 0.056
robust-imp	slot-shift	0.848 $\pm$ 0.067	1.000 $\pm$ 0.000	0.325 $\pm$ 0.001	0.000 $\pm$ 0.000	0.152 $\pm$ 0.067
calib-proto	slot-shift	0.830 $\pm$ 0.109	1.000 $\pm$ 0.000	0.200 $\pm$ 0.000	0.045 $\pm$ 0.037	0.170 $\pm$ 0.109
proto-ID	slot-shift	0.804 $\pm$ 0.087	1.000 $\pm$ 0.000	0.250 $\pm$ 0.000	0.045 $\pm$ 0.037	0.196 $\pm$ 0.087
particle	slot-shift	0.795 $\pm$ 0.072	1.000 $\pm$ 0.000	0.280 $\pm$ 0.000	0.045 $\pm$ 0.037	0.205 $\pm$ 0.072
ensemble	slot-shift	0.750 $\pm$ 0.099	1.000 $\pm$ 0.000	0.314 $\pm$ 0.002	0.009 $\pm$ 0.018	0.250 $\pm$ 0.099
LFI-v5	slot-shift	0.741 $\pm$ 0.083	0.946 $\pm$ 0.044	0.203 $\pm$ 0.021	0.161 $\pm$ 0.074	0.259 $\pm$ 0.083
LFI-v4	slot-shift	0.679 $\pm$ 0.059	0.705 $\pm$ 0.032	0.278 $\pm$ 0.018	0.089 $\pm$ 0.051	0.321 $\pm$ 0.059
visible	slot-shift	0.339 $\pm$ 0.083	0.438 $\pm$ 0.089	0.557 $\pm$ 0.060	0.241 $\pm$ 0.059	0.661 $\pm$ 0.083
threshold	slot-shift	0.339 $\pm$ 0.102	0.438 $\pm$ 0.072	0.659 $\pm$ 0.040	0.312 $\pm$ 0.083	0.661 $\pm$ 0.102

C ALL MAIN PAIRWISE COMPARISONS

Table: All paired comparisons versus LFI-v5

Split	Comparison	Succ diff	CI	Fixture	Param red	Damage red	Wins
adhesive-tether	adaptive-probe	-0.054	0.083	0.009	0.035	-0.018	1/8
adhesive-tether	bayes	0.027	0.106	0.000	0.100	0.036	3/8
adhesive-tether	calib-proto	-0.054	0.091	0.009	0.033	-0.027	1/8
adhesive-tether	ensemble	0.116	0.083	0.009	0.161	0.000	6/8
adhesive-tether	threshold	0.366	0.072	0.420	0.409	0.214	8/8
adhesive-tether	LFI-v4	-0.125	0.083	0.009	-0.056	-0.027	1/8
adhesive-tether	oracle	-0.071	0.070	0.000	-0.172	-0.036	1/8
adhesive-tether	particle	-0.018	0.083	0.000	0.108	0.027	3/8
adhesive-tether	proto-ID	0.045	0.091	0.009	0.084	0.027	3/8
adhesive-tether	rf-probe	-0.027	0.059	0.009	0.077	-0.009	1/8
adhesive-tether	robust-imp	-0.009	0.097	0.009	0.164	-0.036	2/8
adhesive-tether	visible	0.482	0.105	0.607	0.415	0.312	8/8
combined	adaptive-probe	-0.241	0.115	-0.179	-0.055	-0.179	1/8
combined	bayes	-0.143	0.059	-0.116	0.061	-0.125	0/8
combined	calib-proto	-0.232	0.149	-0.196	-0.070	-0.161	1/8
combined	ensemble	-0.062	0.117	-0.196	0.072	-0.107	3/8
combined	threshold	0.179	0.092	0.295	0.350	0.036	7/8
combined	LFI-v4	-0.080	0.072	0.143	0.034	-0.054	1/8
combined	oracle	-0.348	0.104	-0.223	-0.285	-0.196	0/8
combined	particle	-0.116	0.135	-0.116	0.049	-0.080	2/8
combined	proto-ID	-0.196	0.111	-0.196	-0.020	-0.170	1/8
combined	rf-probe	-0.205	0.128	-0.170	-0.006	-0.125	1/8
combined	robust-imp	-0.223	0.089	-0.179	0.077	-0.143	0/8
combined	visible	0.375	0.094	0.571	0.427	0.196	8/8
clamp-hinge	adaptive-probe	-0.241	0.087	-0.232	-0.088	-0.125	0/8
clamp-hinge	bayes	-0.107	0.124	-0.143	0.024	-0.107	2/8
clamp-hinge	calib-proto	-0.179	0.088	-0.232	-0.089	-0.089	1/8
clamp-hinge	ensemble	-0.134	0.077	-0.232	0.031	-0.134	0/8
clamp-hinge	threshold	0.571	0.065	0.688	0.567	0.339	8/8
clamp-hinge	LFI-v4	-0.045	0.079	0.089	0.004	0.027	3/8
clamp-hinge	oracle	-0.268	0.035	-0.232	-0.289	-0.098	0/8
clamp-hinge	particle	-0.143	0.046	-0.196	0.009	-0.054	0/8
clamp-hinge	proto-ID	-0.188	0.091	-0.232	-0.039	-0.071	1/8
clamp-hinge	rf-probe	-0.232	0.074	-0.223	-0.037	-0.089	0/8

Table: All paired comparisons versus LFI-v5 continued

Split	Comparison	Succ diff	CI	Fixture	Param red	Damage red	Wins
clamp-hinge	robust-imp	-0.259	0.064	-0.232	0.041	-0.116	0/8
clamp-hinge	visible	0.348	0.093	0.339	0.274	0.152	8/8
nominal	adaptive-probe	-0.071	0.070	-0.107	-0.025	-0.054	1/8
nominal	bayes	0.027	0.075	-0.098	0.027	-0.027	4/8
nominal	calib-proto	-0.036	0.070	-0.107	-0.023	-0.009	2/8
nominal	ensemble	0.062	0.072	-0.107	0.098	-0.054	5/8
nominal	threshold	0.446	0.044	0.429	0.422	0.241	8/8
nominal	LFI-v4	0.107	0.053	0.161	0.040	0.071	8/8
nominal	oracle	-0.098	0.064	-0.107	-0.223	-0.054	0/8
nominal	particle	-0.045	0.079	-0.107	0.057	-0.018	2/8
nominal	proto-ID	0.027	0.102	-0.107	0.027	-0.018	4/8
nominal	rf-probe	-0.071	0.053	-0.107	0.021	-0.027	1/8
nominal	robust-imp	-0.027	0.064	-0.107	0.094	-0.054	1/8
nominal	visible	0.312	0.106	0.205	0.167	0.098	8/8
slot-shift	adaptive-probe	-0.170	0.127	-0.054	-0.004	-0.143	1/8
slot-shift	bayes	-0.107	0.070	-0.054	0.070	-0.107	0/8
slot-shift	calib-proto	-0.089	0.111	-0.054	-0.003	-0.116	1/8
slot-shift	ensemble	-0.009	0.104	-0.054	0.111	-0.152	5/8
slot-shift	threshold	0.402	0.121	0.509	0.456	0.152	8/8
slot-shift	LFI-v4	0.062	0.093	0.241	0.075	-0.071	5/8
slot-shift	oracle	-0.223	0.125	-0.054	-0.203	-0.143	1/8
slot-shift	particle	-0.054	0.111	-0.054	0.077	-0.116	3/8
slot-shift	proto-ID	-0.062	0.155	-0.054	0.047	-0.116	2/8
slot-shift	rf-probe	-0.125	0.078	-0.054	0.044	-0.098	1/8
slot-shift	robust-imp	-0.107	0.102	-0.054	0.122	-0.161	2/8
slot-shift	visible	0.402	0.102	0.509	0.353	0.080	8/8

D SEED-LEVEL MAIN METRICS

Table: Seed-level main metrics

Method	Split	Seed	Episodes	Success	Fixture	Param	Damage	Repeat
adaptive-probe	adhesive-tether	0	14	0.929	1.000	0.202	0.071	0.071
adaptive-probe	adhesive-tether	1	14	0.786	1.000	0.200	0.000	0.214
adaptive-probe	adhesive-tether	2	14	0.857	1.000	0.203	0.071	0.143
adaptive-probe	adhesive-tether	3	14	0.857	1.000	0.202	0.000	0.143
adaptive-probe	adhesive-tether	4	14	0.857	1.000	0.203	0.000	0.143
adaptive-probe	adhesive-tether	5	14	0.857	0.929	0.237	0.071	0.143
adaptive-probe	adhesive-tether	6	14	0.786	1.000	0.204	0.071	0.214
adaptive-probe	adhesive-tether	7	14	0.929	1.000	0.200	0.000	0.071
bayes	adhesive-tether	0	14	0.786	1.000	0.271	0.143	0.214
bayes	adhesive-tether	1	14	0.714	1.000	0.271	0.071	0.286
bayes	adhesive-tether	2	14	0.929	1.000	0.271	0.143	0.071
bayes	adhesive-tether	3	14	0.929	1.000	0.271	0.071	0.071
bayes	adhesive-tether	4	14	0.714	1.000	0.271	0.071	0.286
bayes	adhesive-tether	5	14	0.643	1.000	0.271	0.071	0.357
bayes	adhesive-tether	6	14	0.643	1.000	0.271	0.000	0.357
bayes	adhesive-tether	7	14	0.857	1.000	0.271	0.143	0.143
calib-proto	adhesive-tether	0	14	0.857	1.000	0.200	0.000	0.143
calib-proto	adhesive-tether	1	14	1.000	1.000	0.200	0.071	0.000
calib-proto	adhesive-tether	2	14	1.000	1.000	0.200	0.000	0.000
calib-proto	adhesive-tether	3	14	0.857	1.000	0.200	0.000	0.143
calib-proto	adhesive-tether	4	14	0.643	1.000	0.200	0.000	0.357
calib-proto	adhesive-tether	5	14	0.857	0.929	0.239	0.000	0.143
calib-proto	adhesive-tether	6	14	0.714	1.000	0.200	0.071	0.286
calib-proto	adhesive-tether	7	14	0.929	1.000	0.200	0.071	0.071
ensemble	adhesive-tether	0	14	0.786	1.000	0.329	0.000	0.214
ensemble	adhesive-tether	1	14	0.571	1.000	0.320	0.071	0.429
ensemble	adhesive-tether	2	14	0.571	1.000	0.329	0.000	0.429
ensemble	adhesive-tether	3	14	0.714	1.000	0.328	0.071	0.286
ensemble	adhesive-tether	4	14	0.571	1.000	0.333	0.000	0.429
ensemble	adhesive-tether	5	14	0.714	0.929	0.366	0.000	0.286
ensemble	adhesive-tether	6	14	0.714	1.000	0.336	0.143	0.286
ensemble	adhesive-tether	7	14	0.857	1.000	0.322	0.143	0.143
threshold	adhesive-tether	0	14	0.500	0.643	0.546	0.214	0.500
threshold	adhesive-tether	1	14	0.429	0.571	0.586	0.286	0.571
threshold	adhesive-tether	2	14	0.429	0.571	0.586	0.357	0.571
threshold	adhesive-tether	3	14	0.429	0.714	0.507	0.143	0.571
threshold	adhesive-tether	4	14	0.286	0.429	0.664	0.500	0.714
threshold	adhesive-tether	5	14	0.286	0.500	0.625	0.214	0.714

Table: Seed-level main metrics continued

Method	Split	Seed	Episodes	Success	Fixture	Param	Damage	Repeat
threshold	adhesive-tether	6	14	0.643	0.643	0.546	0.214	0.357
threshold	adhesive-tether	7	14	0.500	0.571	0.586	0.214	0.500
LFI-v4	adhesive-tether	0	14	0.857	1.000	0.110	0.071	0.143
LFI-v4	adhesive-tether	1	14	1.000	1.000	0.110	0.000	0.000
LFI-v4	adhesive-tether	2	14	1.000	1.000	0.110	0.071	0.000
LFI-v4	adhesive-tether	3	14	1.000	1.000	0.110	0.000	0.000
LFI-v4	adhesive-tether	4	14	0.857	0.929	0.151	0.000	0.143
LFI-v4	adhesive-tether	5	14	0.929	1.000	0.110	0.000	0.071
LFI-v4	adhesive-tether	6	14	0.786	1.000	0.110	0.000	0.214
LFI-v4	adhesive-tether	7	14	1.000	1.000	0.110	0.071	0.000
LFI-v5	adhesive-tether	0	14	0.714	1.000	0.171	0.071	0.286
LFI-v5	adhesive-tether	1	14	0.786	1.000	0.170	0.071	0.214
LFI-v5	adhesive-tether	2	14	0.857	1.000	0.173	0.143	0.143
LFI-v5	adhesive-tether	3	14	0.786	1.000	0.171	0.071	0.214
LFI-v5	adhesive-tether	4	14	0.643	1.000	0.172	0.071	0.357
LFI-v5	adhesive-tether	5	14	0.857	1.000	0.173	0.000	0.143
LFI-v5	adhesive-tether	6	14	0.929	1.000	0.172	0.000	0.071
LFI-v5	adhesive-tether	7	14	0.857	1.000	0.171	0.000	0.143
oracle	adhesive-tether	0	14	0.857	1.000	0.000	0.000	0.143
oracle	adhesive-tether	1	14	0.929	1.000	0.000	0.000	0.071
oracle	adhesive-tether	2	14	0.929	1.000	0.000	0.000	0.071
oracle	adhesive-tether	3	14	0.929	1.000	0.000	0.000	0.071
oracle	adhesive-tether	4	14	0.786	1.000	0.000	0.000	0.214
oracle	adhesive-tether	5	14	0.929	1.000	0.000	0.071	0.071
oracle	adhesive-tether	6	14	0.786	1.000	0.000	0.000	0.214
oracle	adhesive-tether	7	14	0.857	1.000	0.000	0.071	0.143
particle	adhesive-tether	0	14	0.857	1.000	0.280	0.000	0.143
particle	adhesive-tether	1	14	0.714	1.000	0.280	0.071	0.286
particle	adhesive-tether	2	14	0.857	1.000	0.280	0.143	0.143
particle	adhesive-tether	3	14	0.786	1.000	0.280	0.214	0.214
particle	adhesive-tether	4	14	0.857	1.000	0.280	0.071	0.143
particle	adhesive-tether	5	14	0.929	1.000	0.280	0.000	0.071
particle	adhesive-tether	6	14	0.857	1.000	0.280	0.071	0.143
particle	adhesive-tether	7	14	0.714	1.000	0.280	0.071	0.286
proto-ID	adhesive-tether	0	14	0.714	1.000	0.250	0.071	0.286
proto-ID	adhesive-tether	1	14	0.786	1.000	0.250	0.071	0.214
proto-ID	adhesive-tether	2	14	0.643	1.000	0.250	0.214	0.357
proto-ID	adhesive-tether	3	14	0.857	1.000	0.250	0.071	0.143

Table: Seed-level main metrics continued

Method	Split	Seed	Episodes	Success	Fixture	Param	Damage	Repeat
proto-ID	adhesive-tether	4	14	0.786	1.000	0.250	0.071	0.214
proto-ID	adhesive-tether	5	14	0.857	0.929	0.291	0.071	0.143
proto-ID	adhesive-tether	6	14	0.714	1.000	0.250	0.071	0.286
proto-ID	adhesive-tether	7	14	0.714	1.000	0.250	0.000	0.286
rf-probe	adhesive-tether	0	14	0.786	1.000	0.244	0.143	0.214
rf-probe	adhesive-tether	1	14	0.786	1.000	0.241	0.071	0.214
rf-probe	adhesive-tether	2	14	0.857	1.000	0.244	0.000	0.143
rf-probe	adhesive-tether	3	14	0.929	1.000	0.243	0.000	0.071
rf-probe	adhesive-tether	4	14	0.714	1.000	0.245	0.000	0.286
rf-probe	adhesive-tether	5	14	0.714	0.929	0.284	0.000	0.286
rf-probe	adhesive-tether	6	14	1.000	1.000	0.244	0.143	0.000
rf-probe	adhesive-tether	7	14	0.857	1.000	0.242	0.000	0.143
robust-imp	adhesive-tether	0	14	0.857	1.000	0.332	0.000	0.143
robust-imp	adhesive-tether	1	14	0.857	1.000	0.331	0.000	0.143
robust-imp	adhesive-tether	2	14	0.857	1.000	0.333	0.071	0.143
robust-imp	adhesive-tether	3	14	0.786	1.000	0.330	0.000	0.214
robust-imp	adhesive-tether	4	14	0.857	1.000	0.330	0.000	0.143
robust-imp	adhesive-tether	5	14	0.714	0.929	0.367	0.000	0.286
robust-imp	adhesive-tether	6	14	0.714	1.000	0.331	0.071	0.286
robust-imp	adhesive-tether	7	14	0.857	1.000	0.333	0.000	0.143
visible	adhesive-tether	0	14	0.214	0.357	0.611	0.286	0.786
visible	adhesive-tether	1	14	0.429	0.429	0.563	0.500	0.571
visible	adhesive-tether	2	14	0.286	0.357	0.611	0.286	0.714
visible	adhesive-tether	3	14	0.500	0.429	0.563	0.286	0.500
visible	adhesive-tether	4	14	0.357	0.429	0.563	0.429	0.643
visible	adhesive-tether	5	14	0.214	0.500	0.515	0.286	0.786
visible	adhesive-tether	6	14	0.286	0.357	0.611	0.429	0.714
visible	adhesive-tether	7	14	0.286	0.286	0.659	0.429	0.714
adaptive-probe	combined	0	14	0.786	1.000	0.209	0.071	0.214
adaptive-probe	combined	1	14	0.857	1.000	0.206	0.071	0.143
adaptive-probe	combined	2	14	0.857	1.000	0.209	0.000	0.143
adaptive-probe	combined	3	14	0.929	0.929	0.247	0.000	0.071
adaptive-probe	combined	4	14	0.714	1.000	0.207	0.000	0.286
adaptive-probe	combined	5	14	0.714	0.857	0.279	0.000	0.286
adaptive-probe	combined	6	14	0.643	0.929	0.241	0.143	0.357
adaptive-probe	combined	7	14	0.857	0.929	0.240	0.143	0.143
bayes	combined	0	14	0.500	0.857	0.364	0.071	0.500
bayes	combined	1	14	0.714	0.929	0.328	0.000	0.286

Table: Seed-level main metrics continued

Method	Split	Seed	Episodes	Success	Fixture	Param	Damage	Repeat
bayes	combined	2	14	0.714	0.929	0.328	0.143	0.286
bayes	combined	3	14	0.714	0.857	0.364	0.071	0.286
bayes	combined	4	14	0.714	0.929	0.328	0.143	0.286
bayes	combined	5	14	0.714	0.857	0.364	0.143	0.286
bayes	combined	6	14	0.786	0.857	0.364	0.214	0.214
bayes	combined	7	14	0.714	0.929	0.328	0.071	0.286
calib-proto	combined	0	14	1.000	1.000	0.200	0.071	0.000
calib-proto	combined	1	14	0.786	1.000	0.200	0.071	0.214
calib-proto	combined	2	14	0.643	1.000	0.200	0.071	0.357
calib-proto	combined	3	14	0.786	0.929	0.239	0.000	0.214
calib-proto	combined	4	14	0.714	1.000	0.200	0.000	0.286
calib-proto	combined	5	14	0.857	0.929	0.239	0.143	0.143
calib-proto	combined	6	14	0.571	0.929	0.239	0.000	0.429
calib-proto	combined	7	14	0.929	1.000	0.200	0.214	0.071
ensemble	combined	0	14	0.429	1.000	0.342	0.214	0.571
ensemble	combined	1	14	0.714	1.000	0.340	0.143	0.286
ensemble	combined	2	14	0.786	1.000	0.343	0.143	0.214
ensemble	combined	3	14	0.571	0.929	0.390	0.286	0.429
ensemble	combined	4	14	0.714	1.000	0.343	0.000	0.286
ensemble	combined	5	14	0.500	0.929	0.381	0.143	0.500
ensemble	combined	6	14	0.500	1.000	0.345	0.000	0.500
ensemble	combined	7	14	0.714	0.929	0.368	0.071	0.286
threshold	combined	0	14	0.429	0.500	0.625	0.286	0.571
threshold	combined	1	14	0.357	0.357	0.704	0.286	0.643
threshold	combined	2	14	0.286	0.500	0.625	0.357	0.714
threshold	combined	3	14	0.571	0.571	0.586	0.214	0.429
threshold	combined	4	14	0.214	0.429	0.664	0.214	0.786
threshold	combined	5	14	0.357	0.500	0.625	0.143	0.643
threshold	combined	6	14	0.500	0.571	0.586	0.214	0.500
threshold	combined	7	14	0.286	0.429	0.664	0.429	0.714
LFI-v4	combined	0	14	0.714	0.714	0.273	0.214	0.286
LFI-v4	combined	1	14	0.500	0.500	0.395	0.214	0.500
LFI-v4	combined	2	14	0.643	0.643	0.314	0.143	0.357
LFI-v4	combined	3	14	0.643	0.643	0.314	0.143	0.357
LFI-v4	combined	4	14	0.643	0.643	0.314	0.143	0.357
LFI-v4	combined	5	14	0.571	0.571	0.354	0.214	0.429
LFI-v4	combined	6	14	0.714	0.714	0.273	0.143	0.286
LFI-v4	combined	7	14	0.643	0.643	0.314	0.214	0.357

Table: Seed-level main metrics continued

Method	Split	Seed	Episodes	Success	Fixture	Param	Damage	Repeat
LFI-v5	combined	0	14	0.500	0.857	0.245	0.000	0.500
LFI-v5	combined	1	14	0.571	0.714	0.315	0.143	0.429
LFI-v5	combined	2	14	0.571	0.714	0.316	0.143	0.429
LFI-v5	combined	3	14	0.500	0.786	0.281	0.429	0.500
LFI-v5	combined	4	14	0.571	0.786	0.280	0.357	0.429
LFI-v5	combined	5	14	0.571	0.714	0.316	0.357	0.429
LFI-v5	combined	6	14	0.714	0.786	0.279	0.143	0.286
LFI-v5	combined	7	14	0.429	0.857	0.247	0.286	0.571
oracle	combined	0	14	1.000	1.000	0.000	0.071	0.000
oracle	combined	1	14	0.929	1.000	0.000	0.000	0.071
oracle	combined	2	14	1.000	1.000	0.000	0.000	0.000
oracle	combined	3	14	0.786	1.000	0.000	0.071	0.214
oracle	combined	4	14	1.000	1.000	0.000	0.071	0.000
oracle	combined	5	14	0.786	1.000	0.000	0.000	0.214
oracle	combined	6	14	0.786	1.000	0.000	0.000	0.214
oracle	combined	7	14	0.929	1.000	0.000	0.071	0.071
particle	combined	0	14	0.571	0.857	0.351	0.214	0.429
particle	combined	1	14	0.714	0.929	0.316	0.071	0.286
particle	combined	2	14	0.929	1.000	0.280	0.000	0.071
particle	combined	3	14	0.571	0.857	0.351	0.214	0.429
particle	combined	4	14	0.500	0.857	0.351	0.357	0.500
particle	combined	5	14	0.643	0.857	0.351	0.071	0.357
particle	combined	6	14	0.571	0.857	0.351	0.214	0.429
particle	combined	7	14	0.857	0.929	0.316	0.071	0.143
proto-ID	combined	0	14	0.571	1.000	0.250	0.000	0.429
proto-ID	combined	1	14	0.929	1.000	0.250	0.071	0.071
proto-ID	combined	2	14	0.786	1.000	0.250	0.000	0.214
proto-ID	combined	3	14	0.643	0.929	0.291	0.143	0.357
proto-ID	combined	4	14	0.857	1.000	0.250	0.071	0.143
proto-ID	combined	5	14	0.714	0.929	0.291	0.000	0.286
proto-ID	combined	6	14	0.643	0.929	0.291	0.143	0.357
proto-ID	combined	7	14	0.857	1.000	0.250	0.071	0.143
rf-probe	combined	0	14	0.857	0.929	0.289	0.214	0.143
rf-probe	combined	1	14	0.786	1.000	0.249	0.000	0.214
rf-probe	combined	2	14	0.714	1.000	0.249	0.143	0.286
rf-probe	combined	3	14	0.786	0.929	0.290	0.000	0.214
rf-probe	combined	4	14	0.786	1.000	0.251	0.071	0.214
rf-probe	combined	5	14	0.500	0.786	0.370	0.143	0.500

Table: Seed-level main metrics continued

Method	Split	Seed	Episodes	Success	Fixture	Param	Damage	Repeat
rf-probe	combined	6	14	0.714	0.929	0.285	0.071	0.286
rf-probe	combined	7	14	0.929	1.000	0.247	0.214	0.071
robust-imp	combined	0	14	0.714	1.000	0.340	0.000	0.286
robust-imp	combined	1	14	0.929	1.000	0.336	0.143	0.071
robust-imp	combined	2	14	0.857	1.000	0.338	0.143	0.143
robust-imp	combined	3	14	0.714	0.929	0.377	0.071	0.286
robust-imp	combined	4	14	0.929	1.000	0.336	0.000	0.071
robust-imp	combined	5	14	0.643	0.857	0.412	0.143	0.357
robust-imp	combined	6	14	0.714	0.929	0.378	0.071	0.286
robust-imp	combined	7	14	0.714	0.929	0.377	0.143	0.286
visible	combined	0	14	0.214	0.357	0.611	0.429	0.786
visible	combined	1	14	0.214	0.214	0.706	0.429	0.786
visible	combined	2	14	0.000	0.071	0.802	0.286	1.000
visible	combined	3	14	0.286	0.214	0.706	0.643	0.714
visible	combined	4	14	0.286	0.357	0.611	0.286	0.714
visible	combined	5	14	0.143	0.143	0.754	0.500	0.857
visible	combined	6	14	0.143	0.214	0.706	0.429	0.857
visible	combined	7	14	0.143	0.071	0.802	0.429	0.857
adaptive-probe	clamp-hinge	0	14	0.786	1.000	0.201	0.000	0.214
adaptive-probe	clamp-hinge	1	14	0.786	1.000	0.203	0.000	0.214
adaptive-probe	clamp-hinge	2	14	0.857	1.000	0.201	0.000	0.143
adaptive-probe	clamp-hinge	3	14	1.000	1.000	0.200	0.000	0.000
adaptive-probe	clamp-hinge	4	14	0.929	1.000	0.201	0.000	0.071
adaptive-probe	clamp-hinge	5	14	0.929	1.000	0.202	0.071	0.071
adaptive-probe	clamp-hinge	6	14	0.857	1.000	0.200	0.000	0.143
adaptive-probe	clamp-hinge	7	14	0.857	1.000	0.200	0.000	0.143
bayes	clamp-hinge	0	14	0.857	0.929	0.304	0.000	0.143
bayes	clamp-hinge	1	14	0.643	0.786	0.379	0.071	0.357
bayes	clamp-hinge	2	14	0.571	0.857	0.341	0.000	0.429
bayes	clamp-hinge	3	14	1.000	1.000	0.266	0.000	0.000
bayes	clamp-hinge	4	14	0.857	1.000	0.266	0.071	0.143
bayes	clamp-hinge	5	14	0.500	0.929	0.304	0.000	0.500
bayes	clamp-hinge	6	14	0.786	0.857	0.341	0.071	0.214
bayes	clamp-hinge	7	14	0.714	0.929	0.304	0.000	0.286
calib-proto	clamp-hinge	0	14	0.857	1.000	0.200	0.143	0.143
calib-proto	clamp-hinge	1	14	0.786	1.000	0.200	0.000	0.214
calib-proto	clamp-hinge	2	14	0.857	1.000	0.200	0.000	0.143
calib-proto	clamp-hinge	3	14	0.786	1.000	0.200	0.000	0.214

Table: Seed-level main metrics continued

Method	Split	Seed	Episodes	Success	Fixture	Param	Damage	Repeat
calib-proto	clamp-hinge	4	14	0.929	1.000	0.200	0.000	0.071
calib-proto	clamp-hinge	5	14	0.857	1.000	0.200	0.143	0.143
calib-proto	clamp-hinge	6	14	0.786	1.000	0.200	0.071	0.214
calib-proto	clamp-hinge	7	14	0.643	1.000	0.200	0.000	0.357
ensemble	clamp-hinge	0	14	0.786	1.000	0.318	0.000	0.214
ensemble	clamp-hinge	1	14	0.714	1.000	0.327	0.000	0.286
ensemble	clamp-hinge	2	14	0.857	1.000	0.317	0.000	0.143
ensemble	clamp-hinge	3	14	0.571	1.000	0.318	0.000	0.429
ensemble	clamp-hinge	4	14	0.643	1.000	0.320	0.000	0.357
ensemble	clamp-hinge	5	14	0.857	1.000	0.324	0.000	0.143
ensemble	clamp-hinge	6	14	0.929	1.000	0.313	0.000	0.071
ensemble	clamp-hinge	7	14	0.786	1.000	0.323	0.000	0.214
threshold	clamp-hinge	0	14	0.071	0.143	0.821	0.643	0.929
threshold	clamp-hinge	1	14	0.071	0.071	0.861	0.357	0.929
threshold	clamp-hinge	2	14	0.071	0.071	0.861	0.357	0.929
threshold	clamp-hinge	3	14	0.143	0.214	0.782	0.500	0.857
threshold	clamp-hinge	4	14	0.071	0.071	0.861	0.357	0.929
threshold	clamp-hinge	5	14	0.071	0.071	0.861	0.571	0.929
threshold	clamp-hinge	6	14	0.000	0.000	0.900	0.571	1.000
threshold	clamp-hinge	7	14	0.000	0.000	0.900	0.429	1.000
LFI-v4	clamp-hinge	0	14	0.643	0.714	0.273	0.071	0.357
LFI-v4	clamp-hinge	1	14	0.643	0.643	0.314	0.143	0.357
LFI-v4	clamp-hinge	2	14	0.571	0.643	0.314	0.286	0.429
LFI-v4	clamp-hinge	3	14	0.714	0.714	0.273	0.214	0.286
LFI-v4	clamp-hinge	4	14	0.714	0.643	0.314	0.071	0.286
LFI-v4	clamp-hinge	5	14	0.786	0.714	0.273	0.143	0.214
LFI-v4	clamp-hinge	6	14	0.714	0.714	0.273	0.214	0.286
LFI-v4	clamp-hinge	7	14	0.643	0.643	0.314	0.143	0.357
LFI-v5	clamp-hinge	0	14	0.714	0.714	0.315	0.143	0.286
LFI-v5	clamp-hinge	1	14	0.643	0.714	0.315	0.214	0.357
LFI-v5	clamp-hinge	2	14	0.643	0.786	0.281	0.143	0.357
LFI-v5	clamp-hinge	3	14	0.571	0.714	0.314	0.214	0.429
LFI-v5	clamp-hinge	4	14	0.571	0.857	0.247	0.000	0.429
LFI-v5	clamp-hinge	5	14	0.571	0.786	0.282	0.214	0.429
LFI-v5	clamp-hinge	6	14	0.643	0.786	0.280	0.071	0.357
LFI-v5	clamp-hinge	7	14	0.714	0.786	0.280	0.071	0.286
oracle	clamp-hinge	0	14	0.929	1.000	0.000	0.000	0.071
oracle	clamp-hinge	1	14	0.929	1.000	0.000	0.071	0.071

Table: Seed-level main metrics continued

Method	Split	Seed	Episodes	Success	Fixture	Param	Damage	Repeat
oracle	clamp-hinge	2	14	0.929	1.000	0.000	0.000	0.071
oracle	clamp-hinge	3	14	0.857	1.000	0.000	0.000	0.143
oracle	clamp-hinge	4	14	0.857	1.000	0.000	0.071	0.143
oracle	clamp-hinge	5	14	0.786	1.000	0.000	0.000	0.214
oracle	clamp-hinge	6	14	1.000	1.000	0.000	0.071	0.000
oracle	clamp-hinge	7	14	0.929	1.000	0.000	0.071	0.071
particle	clamp-hinge	0	14	0.929	0.929	0.316	0.071	0.071
particle	clamp-hinge	1	14	0.714	0.857	0.351	0.071	0.286
particle	clamp-hinge	2	14	0.786	1.000	0.280	0.071	0.214
particle	clamp-hinge	3	14	0.643	1.000	0.280	0.071	0.357
particle	clamp-hinge	4	14	0.714	1.000	0.280	0.143	0.286
particle	clamp-hinge	5	14	0.786	0.929	0.316	0.071	0.214
particle	clamp-hinge	6	14	0.857	1.000	0.280	0.071	0.143
particle	clamp-hinge	7	14	0.786	1.000	0.280	0.071	0.214
proto-ID	clamp-hinge	0	14	0.929	1.000	0.250	0.000	0.071
proto-ID	clamp-hinge	1	14	0.571	1.000	0.250	0.071	0.429
proto-ID	clamp-hinge	2	14	0.857	1.000	0.250	0.071	0.143
proto-ID	clamp-hinge	3	14	0.857	1.000	0.250	0.071	0.143
proto-ID	clamp-hinge	4	14	0.643	1.000	0.250	0.071	0.357
proto-ID	clamp-hinge	5	14	0.929	1.000	0.250	0.000	0.071
proto-ID	clamp-hinge	6	14	0.857	1.000	0.250	0.071	0.143
proto-ID	clamp-hinge	7	14	0.929	1.000	0.250	0.143	0.071
rf-probe	clamp-hinge	0	14	0.857	1.000	0.248	0.000	0.143
rf-probe	clamp-hinge	1	14	0.929	0.929	0.288	0.143	0.071
rf-probe	clamp-hinge	2	14	1.000	1.000	0.247	0.000	0.000
rf-probe	clamp-hinge	3	14	0.786	1.000	0.248	0.143	0.214
rf-probe	clamp-hinge	4	14	0.857	1.000	0.248	0.000	0.143
rf-probe	clamp-hinge	5	14	0.929	1.000	0.248	0.071	0.071
rf-probe	clamp-hinge	6	14	0.786	1.000	0.248	0.000	0.214
rf-probe	clamp-hinge	7	14	0.786	1.000	0.246	0.000	0.214
robust-imp	clamp-hinge	0	14	1.000	1.000	0.333	0.000	0.000
robust-imp	clamp-hinge	1	14	0.786	1.000	0.330	0.071	0.214
robust-imp	clamp-hinge	2	14	0.857	1.000	0.331	0.000	0.143
robust-imp	clamp-hinge	3	14	0.857	1.000	0.330	0.000	0.143
robust-imp	clamp-hinge	4	14	1.000	1.000	0.331	0.000	0.000
robust-imp	clamp-hinge	5	14	0.857	1.000	0.329	0.000	0.143
robust-imp	clamp-hinge	6	14	0.929	1.000	0.330	0.071	0.071
robust-imp	clamp-hinge	7	14	0.857	1.000	0.329	0.000	0.143

Table: Seed-level main metrics continued

Method	Split	Seed	Episodes	Success	Fixture	Param	Damage	Repeat
visible	clamp-hinge	0	14	0.214	0.286	0.659	0.214	0.786
visible	clamp-hinge	1	14	0.214	0.429	0.563	0.214	0.786
visible	clamp-hinge	2	14	0.357	0.429	0.563	0.286	0.643
visible	clamp-hinge	3	14	0.357	0.500	0.515	0.500	0.643
visible	clamp-hinge	4	14	0.357	0.357	0.611	0.286	0.643
visible	clamp-hinge	5	14	0.286	0.429	0.563	0.143	0.714
visible	clamp-hinge	6	14	0.357	0.500	0.515	0.429	0.643
visible	clamp-hinge	7	14	0.143	0.500	0.515	0.214	0.857
adaptive-probe	nominal	0	14	0.857	1.000	0.198	0.000	0.143
adaptive-probe	nominal	1	14	1.000	1.000	0.197	0.000	0.000
adaptive-probe	nominal	2	14	0.929	1.000	0.198	0.000	0.071
adaptive-probe	nominal	3	14	0.857	1.000	0.199	0.000	0.143
adaptive-probe	nominal	4	14	0.786	1.000	0.199	0.000	0.214
adaptive-probe	nominal	5	14	0.857	1.000	0.199	0.071	0.143
adaptive-probe	nominal	6	14	0.929	1.000	0.196	0.000	0.071
adaptive-probe	nominal	7	14	0.929	1.000	0.196	0.000	0.071
bayes	nominal	0	14	0.786	1.000	0.245	0.000	0.214
bayes	nominal	1	14	0.714	1.000	0.245	0.071	0.286
bayes	nominal	2	14	0.929	1.000	0.245	0.071	0.071
bayes	nominal	3	14	0.786	0.929	0.284	0.071	0.214
bayes	nominal	4	14	0.786	1.000	0.245	0.071	0.214
bayes	nominal	5	14	0.857	1.000	0.245	0.000	0.143
bayes	nominal	6	14	0.786	1.000	0.245	0.000	0.214
bayes	nominal	7	14	0.714	1.000	0.245	0.000	0.286
calib-proto	nominal	0	14	0.857	1.000	0.200	0.000	0.143
calib-proto	nominal	1	14	0.929	1.000	0.200	0.071	0.071
calib-proto	nominal	2	14	0.786	1.000	0.200	0.000	0.214
calib-proto	nominal	3	14	0.929	1.000	0.200	0.071	0.071
calib-proto	nominal	4	14	0.857	1.000	0.200	0.071	0.143
calib-proto	nominal	5	14	0.786	1.000	0.200	0.071	0.214
calib-proto	nominal	6	14	0.857	1.000	0.200	0.071	0.143
calib-proto	nominal	7	14	0.857	1.000	0.200	0.071	0.143
ensemble	nominal	0	14	0.786	1.000	0.316	0.000	0.214
ensemble	nominal	1	14	0.643	1.000	0.320	0.000	0.357
ensemble	nominal	2	14	0.643	1.000	0.324	0.000	0.357
ensemble	nominal	3	14	0.714	1.000	0.321	0.000	0.286
ensemble	nominal	4	14	0.714	1.000	0.330	0.000	0.286
ensemble	nominal	5	14	0.857	1.000	0.324	0.071	0.143

Table: Seed-level main metrics continued

Method	Split	Seed	Episodes	Success	Fixture	Param	Damage	Repeat
ensemble	nominal	6	14	0.929	1.000	0.315	0.000	0.071
ensemble	nominal	7	14	0.786	1.000	0.317	0.000	0.214
threshold	nominal	0	14	0.357	0.429	0.664	0.500	0.643
threshold	nominal	1	14	0.357	0.429	0.664	0.286	0.643
threshold	nominal	2	14	0.429	0.500	0.625	0.500	0.571
threshold	nominal	3	14	0.357	0.429	0.664	0.214	0.643
threshold	nominal	4	14	0.429	0.429	0.664	0.143	0.571
threshold	nominal	5	14	0.357	0.571	0.586	0.357	0.643
threshold	nominal	6	14	0.286	0.429	0.664	0.286	0.714
threshold	nominal	7	14	0.429	0.500	0.625	0.143	0.571
LFI-v4	nominal	0	14	0.714	0.714	0.273	0.214	0.286
LFI-v4	nominal	1	14	0.714	0.714	0.273	0.143	0.286
LFI-v4	nominal	2	14	0.714	0.714	0.273	0.071	0.286
LFI-v4	nominal	3	14	0.643	0.714	0.273	0.214	0.357
LFI-v4	nominal	4	14	0.786	0.857	0.191	0.071	0.214
LFI-v4	nominal	5	14	0.786	0.786	0.232	0.143	0.214
LFI-v4	nominal	6	14	0.714	0.714	0.273	0.143	0.286
LFI-v4	nominal	7	14	0.643	0.643	0.314	0.071	0.357
LFI-v5	nominal	0	14	0.857	0.857	0.242	0.143	0.143
LFI-v5	nominal	1	14	0.786	0.857	0.240	0.071	0.214
LFI-v5	nominal	2	14	0.786	0.857	0.238	0.071	0.214
LFI-v5	nominal	3	14	0.714	0.786	0.275	0.071	0.286
LFI-v5	nominal	4	14	0.857	1.000	0.171	0.071	0.143
LFI-v5	nominal	5	14	0.857	0.929	0.205	0.071	0.143
LFI-v5	nominal	6	14	0.786	0.857	0.240	0.000	0.214
LFI-v5	nominal	7	14	0.929	1.000	0.170	0.000	0.071
oracle	nominal	0	14	1.000	1.000	0.000	0.000	0.000
oracle	nominal	1	14	0.857	1.000	0.000	0.000	0.143
oracle	nominal	2	14	0.857	1.000	0.000	0.000	0.143
oracle	nominal	3	14	1.000	1.000	0.000	0.071	0.000
oracle	nominal	4	14	0.857	1.000	0.000	0.000	0.143
oracle	nominal	5	14	0.929	1.000	0.000	0.000	0.071
oracle	nominal	6	14	0.929	1.000	0.000	0.000	0.071
oracle	nominal	7	14	0.929	1.000	0.000	0.000	0.071
particle	nominal	0	14	0.857	1.000	0.280	0.143	0.143
particle	nominal	1	14	1.000	1.000	0.280	0.000	0.000
particle	nominal	2	14	0.714	1.000	0.280	0.071	0.286
particle	nominal	3	14	0.929	1.000	0.280	0.000	0.071

Table: Seed-level main metrics continued

Method	Split	Seed	Episodes	Success	Fixture	Param	Damage	Repeat
particle	nominal	4	14	0.786	1.000	0.280	0.000	0.214
particle	nominal	5	14	0.929	1.000	0.280	0.143	0.071
particle	nominal	6	14	0.786	1.000	0.280	0.000	0.214
particle	nominal	7	14	0.929	1.000	0.280	0.000	0.071
proto-ID	nominal	0	14	0.786	1.000	0.250	0.071	0.214
proto-ID	nominal	1	14	0.643	1.000	0.250	0.143	0.357
proto-ID	nominal	2	14	0.929	1.000	0.250	0.000	0.071
proto-ID	nominal	3	14	0.857	1.000	0.250	0.000	0.143
proto-ID	nominal	4	14	0.786	1.000	0.250	0.071	0.214
proto-ID	nominal	5	14	0.571	1.000	0.250	0.000	0.429
proto-ID	nominal	6	14	0.857	1.000	0.250	0.000	0.143
proto-ID	nominal	7	14	0.929	1.000	0.250	0.071	0.071
rf-probe	nominal	0	14	0.929	1.000	0.245	0.000	0.071
rf-probe	nominal	1	14	0.929	1.000	0.244	0.000	0.071
rf-probe	nominal	2	14	0.857	1.000	0.242	0.071	0.143
rf-probe	nominal	3	14	0.857	1.000	0.245	0.071	0.143
rf-probe	nominal	4	14	0.929	1.000	0.244	0.143	0.071
rf-probe	nominal	5	14	0.857	1.000	0.245	0.000	0.143
rf-probe	nominal	6	14	0.929	1.000	0.244	0.000	0.071
rf-probe	nominal	7	14	0.857	1.000	0.243	0.000	0.143
robust-imp	nominal	0	14	0.857	1.000	0.320	0.000	0.143
robust-imp	nominal	1	14	0.929	1.000	0.316	0.000	0.071
robust-imp	nominal	2	14	0.857	1.000	0.318	0.000	0.143
robust-imp	nominal	3	14	0.857	1.000	0.315	0.000	0.143
robust-imp	nominal	4	14	0.857	1.000	0.319	0.000	0.143
robust-imp	nominal	5	14	0.857	1.000	0.316	0.071	0.143
robust-imp	nominal	6	14	0.786	1.000	0.315	0.000	0.214
robust-imp	nominal	7	14	0.786	1.000	0.317	0.000	0.214
visible	nominal	0	14	0.214	0.429	0.563	0.286	0.786
visible	nominal	1	14	0.500	0.714	0.371	0.214	0.500
visible	nominal	2	14	0.429	0.643	0.419	0.000	0.571
visible	nominal	3	14	0.571	0.571	0.467	0.071	0.429
visible	nominal	4	14	0.643	0.786	0.324	0.357	0.357
visible	nominal	5	14	0.571	0.714	0.371	0.071	0.429
visible	nominal	6	14	0.571	0.714	0.371	0.214	0.429
visible	nominal	7	14	0.571	0.929	0.228	0.071	0.429
adaptive-probe	slot-shift	0	14	1.000	1.000	0.201	0.000	0.000
adaptive-probe	slot-shift	1	14	0.929	1.000	0.199	0.000	0.071

Table: Seed-level main metrics continued

Method	Split	Seed	Episodes	Success	Fixture	Param	Damage	Repeat
adaptive-probe	slot-shift	2	14	0.857	1.000	0.200	0.071	0.143
adaptive-probe	slot-shift	3	14	0.857	1.000	0.199	0.000	0.143
adaptive-probe	slot-shift	4	14	1.000	1.000	0.199	0.000	0.000
adaptive-probe	slot-shift	5	14	0.786	1.000	0.199	0.000	0.214
adaptive-probe	slot-shift	6	14	1.000	1.000	0.198	0.000	0.000
adaptive-probe	slot-shift	7	14	0.857	1.000	0.199	0.071	0.143
bayes	slot-shift	0	14	0.786	1.000	0.274	0.071	0.214
bayes	slot-shift	1	14	0.714	1.000	0.274	0.000	0.286
bayes	slot-shift	2	14	0.929	1.000	0.274	0.071	0.071
bayes	slot-shift	3	14	0.929	1.000	0.274	0.000	0.071
bayes	slot-shift	4	14	0.857	1.000	0.274	0.071	0.143
bayes	slot-shift	5	14	0.929	1.000	0.274	0.143	0.071
bayes	slot-shift	6	14	0.786	1.000	0.274	0.071	0.214
bayes	slot-shift	7	14	0.857	1.000	0.274	0.000	0.143
calib-proto	slot-shift	0	14	0.500	1.000	0.200	0.000	0.500
calib-proto	slot-shift	1	14	0.714	1.000	0.200	0.143	0.286
calib-proto	slot-shift	2	14	0.857	1.000	0.200	0.000	0.143
calib-proto	slot-shift	3	14	1.000	1.000	0.200	0.071	0.000
calib-proto	slot-shift	4	14	0.929	1.000	0.200	0.000	0.071
calib-proto	slot-shift	5	14	0.929	1.000	0.200	0.000	0.071
calib-proto	slot-shift	6	14	0.857	1.000	0.200	0.071	0.143
calib-proto	slot-shift	7	14	0.857	1.000	0.200	0.071	0.143
ensemble	slot-shift	0	14	0.714	1.000	0.316	0.000	0.286
ensemble	slot-shift	1	14	0.571	1.000	0.315	0.000	0.429
ensemble	slot-shift	2	14	0.571	1.000	0.318	0.000	0.429
ensemble	slot-shift	3	14	0.714	1.000	0.313	0.000	0.286
ensemble	slot-shift	4	14	0.714	1.000	0.310	0.000	0.286
ensemble	slot-shift	5	14	0.857	1.000	0.315	0.071	0.143
ensemble	slot-shift	6	14	0.929	1.000	0.313	0.000	0.071
ensemble	slot-shift	7	14	0.929	1.000	0.314	0.000	0.071
threshold	slot-shift	0	14	0.286	0.429	0.664	0.286	0.714
threshold	slot-shift	1	14	0.143	0.429	0.664	0.357	0.857
threshold	slot-shift	2	14	0.429	0.429	0.664	0.214	0.571
threshold	slot-shift	3	14	0.643	0.643	0.546	0.214	0.357
threshold	slot-shift	4	14	0.357	0.429	0.664	0.357	0.643
threshold	slot-shift	5	14	0.286	0.500	0.625	0.286	0.714
threshold	slot-shift	6	14	0.286	0.357	0.704	0.571	0.714
threshold	slot-shift	7	14	0.286	0.286	0.743	0.214	0.714

Table: Seed-level main metrics continued

Method	Split	Seed	Episodes	Success	Fixture	Param	Damage	Repeat
LFI-v4	slot-shift	0	14	0.571	0.643	0.314	0.000	0.429
LFI-v4	slot-shift	1	14	0.571	0.714	0.273	0.071	0.429
LFI-v4	slot-shift	2	14	0.786	0.714	0.273	0.071	0.214
LFI-v4	slot-shift	3	14	0.786	0.786	0.232	0.071	0.214
LFI-v4	slot-shift	4	14	0.714	0.714	0.273	0.143	0.286
LFI-v4	slot-shift	5	14	0.643	0.714	0.273	0.000	0.357
LFI-v4	slot-shift	6	14	0.643	0.643	0.314	0.143	0.357
LFI-v4	slot-shift	7	14	0.714	0.714	0.273	0.214	0.286
LFI-v5	slot-shift	0	14	0.714	0.857	0.247	0.357	0.286
LFI-v5	slot-shift	1	14	0.643	0.929	0.211	0.143	0.357
LFI-v5	slot-shift	2	14	0.643	0.857	0.246	0.143	0.357
LFI-v5	slot-shift	3	14	0.786	0.929	0.212	0.071	0.214
LFI-v5	slot-shift	4	14	0.786	1.000	0.179	0.000	0.214
LFI-v5	slot-shift	5	14	0.929	1.000	0.177	0.143	0.071
LFI-v5	slot-shift	6	14	0.571	1.000	0.177	0.214	0.429
LFI-v5	slot-shift	7	14	0.857	1.000	0.178	0.214	0.143
oracle	slot-shift	0	14	1.000	1.000	0.000	0.000	0.000
oracle	slot-shift	1	14	1.000	1.000	0.000	0.000	0.000
oracle	slot-shift	2	14	1.000	1.000	0.000	0.071	0.000
oracle	slot-shift	3	14	1.000	1.000	0.000	0.000	0.000
oracle	slot-shift	4	14	0.929	1.000	0.000	0.000	0.071
oracle	slot-shift	5	14	0.786	1.000	0.000	0.000	0.214
oracle	slot-shift	6	14	1.000	1.000	0.000	0.000	0.000
oracle	slot-shift	7	14	1.000	1.000	0.000	0.071	0.000
particle	slot-shift	0	14	0.643	1.000	0.280	0.000	0.357
particle	slot-shift	1	14	0.929	1.000	0.280	0.000	0.071
particle	slot-shift	2	14	0.857	1.000	0.280	0.071	0.143
particle	slot-shift	3	14	0.643	1.000	0.280	0.000	0.357
particle	slot-shift	4	14	0.786	1.000	0.280	0.071	0.214
particle	slot-shift	5	14	0.857	1.000	0.280	0.143	0.143
particle	slot-shift	6	14	0.786	1.000	0.280	0.071	0.214
particle	slot-shift	7	14	0.857	1.000	0.280	0.000	0.143
proto-ID	slot-shift	0	14	0.714	1.000	0.250	0.071	0.286
proto-ID	slot-shift	1	14	0.857	1.000	0.250	0.143	0.143
proto-ID	slot-shift	2	14	0.929	1.000	0.250	0.000	0.071
proto-ID	slot-shift	3	14	0.857	1.000	0.250	0.000	0.143
proto-ID	slot-shift	4	14	0.929	1.000	0.250	0.071	0.071
proto-ID	slot-shift	5	14	0.571	1.000	0.250	0.000	0.429

Table: Seed-level main metrics continued

Method	Split	Seed	Episodes	Success	Fixture	Param	Damage	Repeat
proto-ID	slot-shift	6	14	0.857	1.000	0.250	0.071	0.143
proto-ID	slot-shift	7	14	0.714	1.000	0.250	0.000	0.286
rf-probe	slot-shift	0	14	0.929	1.000	0.248	0.143	0.071
rf-probe	slot-shift	1	14	0.857	1.000	0.246	0.000	0.143
rf-probe	slot-shift	2	14	0.857	1.000	0.247	0.071	0.143
rf-probe	slot-shift	3	14	0.857	1.000	0.248	0.071	0.143
rf-probe	slot-shift	4	14	0.929	1.000	0.247	0.143	0.071
rf-probe	slot-shift	5	14	0.857	1.000	0.247	0.000	0.143
rf-probe	slot-shift	6	14	0.786	1.000	0.247	0.000	0.214
rf-probe	slot-shift	7	14	0.857	1.000	0.247	0.071	0.143
robust-imp	slot-shift	0	14	0.857	1.000	0.323	0.000	0.143
robust-imp	slot-shift	1	14	0.714	1.000	0.326	0.000	0.286
robust-imp	slot-shift	2	14	0.929	1.000	0.324	0.000	0.071
robust-imp	slot-shift	3	14	0.714	1.000	0.329	0.000	0.286
robust-imp	slot-shift	4	14	1.000	1.000	0.323	0.000	0.000
robust-imp	slot-shift	5	14	0.857	1.000	0.324	0.000	0.143
robust-imp	slot-shift	6	14	0.857	1.000	0.325	0.000	0.143
robust-imp	slot-shift	7	14	0.857	1.000	0.327	0.000	0.143
visible	slot-shift	0	14	0.214	0.214	0.706	0.286	0.786
visible	slot-shift	1	14	0.500	0.500	0.515	0.214	0.500
visible	slot-shift	2	14	0.429	0.500	0.515	0.214	0.571
visible	slot-shift	3	14	0.357	0.357	0.611	0.357	0.643
visible	slot-shift	4	14	0.357	0.357	0.611	0.357	0.643
visible	slot-shift	5	14	0.429	0.500	0.515	0.143	0.571
visible	slot-shift	6	14	0.143	0.643	0.419	0.143	0.857
visible	slot-shift	7	14	0.286	0.429	0.563	0.214	0.714

E SEED-LEVEL AGGREGATE METRICS

Table: Seed-level aggregate metrics

Method	Seed	Episodes	Success	Fixture	Param	Damage	Repeat
adaptive-probe	0	56	0.875	1.000	0.203	0.036	0.125
adaptive-probe	1	56	0.839	1.000	0.202	0.018	0.161
adaptive-probe	2	56	0.857	1.000	0.203	0.036	0.143
adaptive-probe	3	56	0.911	0.982	0.212	0.000	0.089
adaptive-probe	4	56	0.875	1.000	0.202	0.000	0.125
adaptive-probe	5	56	0.821	0.946	0.229	0.036	0.179
adaptive-probe	6	56	0.821	0.982	0.211	0.054	0.179
adaptive-probe	7	56	0.875	0.982	0.210	0.054	0.125
bayes	0	56	0.732	0.946	0.303	0.071	0.268
bayes	1	56	0.696	0.929	0.313	0.036	0.304
bayes	2	56	0.786	0.946	0.304	0.089	0.214
bayes	3	56	0.893	0.964	0.294	0.036	0.107
bayes	4	56	0.786	0.982	0.285	0.089	0.214
bayes	5	56	0.696	0.946	0.303	0.089	0.304
bayes	6	56	0.750	0.929	0.312	0.089	0.250
bayes	7	56	0.786	0.964	0.294	0.054	0.214
calib-proto	0	56	0.804	1.000	0.200	0.054	0.196
calib-proto	1	56	0.821	1.000	0.200	0.071	0.179
calib-proto	2	56	0.839	1.000	0.200	0.018	0.161
calib-proto	3	56	0.857	0.982	0.210	0.018	0.143
calib-proto	4	56	0.804	1.000	0.200	0.000	0.196
calib-proto	5	56	0.875	0.964	0.219	0.071	0.125
calib-proto	6	56	0.732	0.982	0.210	0.054	0.268
calib-proto	7	56	0.839	1.000	0.200	0.089	0.161
ensemble	0	56	0.679	1.000	0.326	0.054	0.321
ensemble	1	56	0.643	1.000	0.326	0.054	0.357
ensemble	2	56	0.696	1.000	0.327	0.036	0.304
ensemble	3	56	0.643	0.982	0.337	0.089	0.357
ensemble	4	56	0.661	1.000	0.327	0.000	0.339
ensemble	5	56	0.732	0.964	0.346	0.054	0.268
ensemble	6	56	0.768	1.000	0.327	0.036	0.232
ensemble	7	56	0.821	0.982	0.332	0.054	0.179
threshold	0	56	0.321	0.429	0.664	0.357	0.679
threshold	1	56	0.250	0.357	0.704	0.321	0.750
threshold	2	56	0.304	0.393	0.684	0.321	0.696
threshold	3	56	0.446	0.536	0.605	0.268	0.554
threshold	4	56	0.232	0.339	0.713	0.357	0.768
threshold	5	56	0.250	0.393	0.684	0.304	0.750

Table: Seed-level aggregate metrics continued

Method	Seed	Episodes	Success	Fixture	Param	Damage	Repeat
threshold	6	56	0.357	0.393	0.684	0.393	0.643
threshold	7	56	0.268	0.321	0.723	0.321	0.732
LFI-v4	0	56	0.696	0.768	0.242	0.089	0.304
LFI-v4	1	56	0.679	0.714	0.273	0.107	0.321
LFI-v4	2	56	0.750	0.750	0.253	0.143	0.250
LFI-v4	3	56	0.786	0.786	0.232	0.107	0.214
LFI-v4	4	56	0.732	0.732	0.263	0.089	0.268
LFI-v4	5	56	0.732	0.750	0.253	0.089	0.268
LFI-v4	6	56	0.714	0.768	0.242	0.125	0.286
LFI-v4	7	56	0.750	0.750	0.253	0.161	0.250
LFI-v5	0	56	0.661	0.857	0.244	0.143	0.339
LFI-v5	1	56	0.661	0.839	0.253	0.143	0.339
LFI-v5	2	56	0.679	0.839	0.254	0.143	0.321
LFI-v5	3	56	0.661	0.857	0.245	0.196	0.339
LFI-v5	4	56	0.643	0.911	0.220	0.107	0.357
LFI-v5	5	56	0.732	0.875	0.237	0.179	0.268
LFI-v5	6	56	0.714	0.893	0.227	0.107	0.286
LFI-v5	7	56	0.714	0.911	0.219	0.143	0.286
oracle	0	56	0.946	1.000	0.000	0.018	0.054
oracle	1	56	0.946	1.000	0.000	0.018	0.054
oracle	2	56	0.964	1.000	0.000	0.018	0.036
oracle	3	56	0.893	1.000	0.000	0.018	0.107
oracle	4	56	0.893	1.000	0.000	0.036	0.107
oracle	5	56	0.821	1.000	0.000	0.018	0.179
oracle	6	56	0.893	1.000	0.000	0.018	0.107
oracle	7	56	0.929	1.000	0.000	0.071	0.071
particle	0	56	0.750	0.946	0.307	0.071	0.250
particle	1	56	0.768	0.946	0.307	0.054	0.232
particle	2	56	0.857	1.000	0.280	0.071	0.143
particle	3	56	0.661	0.964	0.298	0.125	0.339
particle	4	56	0.714	0.964	0.298	0.161	0.286
particle	5	56	0.804	0.946	0.307	0.071	0.196
particle	6	56	0.768	0.964	0.298	0.107	0.232
particle	7	56	0.804	0.982	0.289	0.054	0.196
proto-ID	0	56	0.732	1.000	0.250	0.036	0.268
proto-ID	1	56	0.786	1.000	0.250	0.089	0.214
proto-ID	2	56	0.804	1.000	0.250	0.071	0.196
proto-ID	3	56	0.804	0.982	0.260	0.071	0.196

Table: Seed-level aggregate metrics continued

Method	Seed	Episodes	Success	Fixture	Param	Damage	Repeat
proto-ID	4	56	0.804	1.000	0.250	0.071	0.196
proto-ID	5	56	0.768	0.964	0.270	0.018	0.232
proto-ID	6	56	0.768	0.982	0.260	0.089	0.232
proto-ID	7	56	0.804	1.000	0.250	0.054	0.196
rf-probe	0	56	0.857	0.982	0.257	0.125	0.143
rf-probe	1	56	0.839	0.982	0.256	0.054	0.161
rf-probe	2	56	0.857	1.000	0.247	0.054	0.143
rf-probe	3	56	0.839	0.982	0.257	0.054	0.161
rf-probe	4	56	0.821	1.000	0.248	0.054	0.179
rf-probe	5	56	0.750	0.929	0.287	0.054	0.250
rf-probe	6	56	0.821	0.982	0.256	0.054	0.179
rf-probe	7	56	0.857	1.000	0.246	0.071	0.143
robust-imp	0	56	0.857	1.000	0.332	0.000	0.143
robust-imp	1	56	0.821	1.000	0.331	0.054	0.179
robust-imp	2	56	0.875	1.000	0.331	0.054	0.125
robust-imp	3	56	0.768	0.982	0.341	0.018	0.232
robust-imp	4	56	0.946	1.000	0.330	0.000	0.054
robust-imp	5	56	0.768	0.946	0.358	0.036	0.232
robust-imp	6	56	0.804	0.982	0.341	0.054	0.196
robust-imp	7	56	0.821	0.982	0.341	0.036	0.179
visible	0	56	0.214	0.304	0.647	0.304	0.786
visible	1	56	0.339	0.393	0.587	0.339	0.661
visible	2	56	0.268	0.339	0.623	0.268	0.732
visible	3	56	0.375	0.375	0.599	0.446	0.625
visible	4	56	0.339	0.375	0.599	0.339	0.661
visible	5	56	0.268	0.393	0.587	0.268	0.732
visible	6	56	0.232	0.429	0.563	0.357	0.768
visible	7	56	0.214	0.321	0.635	0.321	0.786

F SEED-LEVEL ABLATION METRICS

Table: Seed-level ablation metrics

Variant	Seed	Episodes	Success	Fixture	Param	Damage	Repeat
full-v5	0	10	0.500	0.800	0.273	0.400	0.500
full-v5	1	10	0.700	0.700	0.322	0.000	0.300
full-v5	2	10	0.600	0.800	0.271	0.300	0.400
full-v5	3	10	0.500	0.800	0.273	0.400	0.500
full-v5	4	10	0.600	0.700	0.324	0.200	0.400
full-v5	5	10	0.500	0.800	0.276	0.500	0.500
full-v5	6	10	0.700	0.800	0.273	0.500	0.300
full-v5	7	10	0.400	0.700	0.323	0.300	0.600
no-adapt	0	10	0.800	0.800	0.271	0.100	0.200
no-adapt	1	10	0.700	0.800	0.273	0.000	0.300
no-adapt	2	10	0.500	0.800	0.270	0.000	0.500
no-adapt	3	10	0.900	0.900	0.224	0.000	0.100
no-adapt	4	10	0.700	0.700	0.322	0.000	0.300
no-adapt	5	10	0.700	0.800	0.274	0.100	0.300
no-adapt	6	10	0.700	0.900	0.223	0.100	0.300
no-adapt	7	10	0.600	0.800	0.274	0.100	0.400
no-calib	0	10	0.500	0.800	0.319	0.000	0.500
no-calib	1	10	0.800	0.800	0.321	0.000	0.200
no-calib	2	10	0.600	0.800	0.317	0.100	0.400
no-calib	3	10	0.600	0.800	0.319	0.400	0.400
no-calib	4	10	0.500	0.700	0.371	0.200	0.500
no-calib	5	10	0.800	0.800	0.322	0.000	0.200
no-calib	6	10	0.900	0.900	0.271	0.100	0.100
no-calib	7	10	0.600	0.700	0.370	0.100	0.400
no-aniso	0	10	0.400	0.700	0.391	0.300	0.600
no-aniso	1	10	0.300	0.600	0.442	0.300	0.700
no-aniso	2	10	0.400	0.700	0.390	0.000	0.600
no-aniso	3	10	0.400	0.800	0.344	0.200	0.600
no-aniso	4	10	0.400	0.600	0.443	0.400	0.600
no-aniso	5	10	0.600	0.800	0.347	0.300	0.400
no-aniso	6	10	0.600	0.700	0.391	0.300	0.400
no-aniso	7	10	0.400	0.600	0.442	0.200	0.600
no-hysteresis	0	10	0.600	0.600	0.439	0.400	0.400
no-hysteresis	1	10	0.500	0.700	0.392	0.300	0.500
no-hysteresis	2	10	0.500	0.600	0.437	0.100	0.500
no-hysteresis	3	10	0.300	0.600	0.439	0.300	0.700
no-hysteresis	4	10	0.600	0.600	0.441	0.400	0.400
no-hysteresis	5	10	0.300	0.500	0.488	0.000	0.700

Table: Seed-level ablation metrics continued

Variant	Seed	Episodes	Success	Fixture	Param	Damage	Repeat
no-hysteresis	6	10	0.500	0.600	0.439	0.200	0.500
no-hysteresis	7	10	0.600	0.600	0.441	0.400	0.400
no-particle	0	10	0.400	0.400	0.466	0.100	0.600
no-particle	1	10	0.200	0.200	0.564	0.400	0.800
no-particle	2	10	0.300	0.500	0.419	0.100	0.700
no-particle	3	10	0.500	0.400	0.466	0.100	0.500
no-particle	4	10	0.400	0.400	0.469	0.500	0.600
no-particle	5	10	0.300	0.300	0.514	0.200	0.700
no-particle	6	10	0.400	0.500	0.418	0.200	0.600
no-particle	7	10	0.300	0.300	0.516	0.200	0.700
no-release	0	10	0.500	0.700	0.395	0.500	0.500
no-release	1	10	0.700	0.800	0.345	0.300	0.300
no-release	2	10	0.500	0.600	0.441	0.100	0.500
no-release	3	10	0.300	0.700	0.394	0.100	0.700
no-release	4	10	0.600	0.800	0.347	0.600	0.400
no-release	5	10	0.400	0.600	0.443	0.400	0.600
no-release	6	10	0.600	0.700	0.393	0.000	0.400
no-release	7	10	0.500	0.800	0.347	0.100	0.500
no-margin	0	10	0.300	0.800	0.273	0.500	0.700
no-margin	1	10	0.400	0.700	0.322	0.500	0.600
no-margin	2	10	0.500	0.800	0.271	0.400	0.500
no-margin	3	10	0.700	0.800	0.273	0.300	0.300
no-margin	4	10	0.600	0.700	0.324	0.500	0.400
no-margin	5	10	0.500	0.800	0.276	0.200	0.500
no-margin	6	10	0.600	0.800	0.273	0.300	0.400
no-margin	7	10	0.400	0.700	0.323	0.300	0.600
no-torque	0	10	0.400	0.500	0.491	0.500	0.600
no-torque	1	10	0.100	0.100	0.681	0.300	0.900
no-torque	2	10	0.200	0.400	0.538	0.100	0.800
no-torque	3	10	0.400	0.400	0.538	0.500	0.600
no-torque	4	10	0.300	0.400	0.540	0.500	0.700
no-torque	5	10	0.300	0.400	0.539	0.400	0.700
no-torque	6	10	0.300	0.600	0.443	0.300	0.700
no-torque	7	10	0.400	0.400	0.540	0.400	0.600

G SEED-LEVEL FIXED-RISK METRICS

Table: Seed-level fixed-risk metrics

Method	Budget	Seed	Episodes	Success	Fixture	Damage	Repeat
adaptive-probe	0.08	0	8	0.750	0.875	0.000	0.250
adaptive-probe	0.08	1	8	0.875	1.000	0.125	0.000
adaptive-probe	0.08	2	8	0.750	1.000	0.000	0.250
adaptive-probe	0.08	3	8	0.625	1.000	0.125	0.250
adaptive-probe	0.08	4	8	0.500	1.000	0.125	0.500
adaptive-probe	0.08	5	8	0.750	0.875	0.125	0.250
adaptive-probe	0.08	6	8	0.750	0.875	0.125	0.250
adaptive-probe	0.08	7	8	0.625	1.000	0.000	0.375
bayes	0.08	0	8	0.375	0.750	0.125	0.375
bayes	0.08	1	8	0.625	1.000	0.000	0.375
bayes	0.08	2	8	0.625	0.875	0.125	0.250
bayes	0.08	3	8	0.500	0.875	0.125	0.375
bayes	0.08	4	8	0.625	0.875	0.125	0.250
bayes	0.08	5	8	0.250	0.750	0.250	0.625
bayes	0.08	6	8	0.125	0.750	0.375	0.875
bayes	0.08	7	8	0.500	0.875	0.375	0.250
calib-proto	0.08	0	8	0.250	0.875	0.250	0.500
calib-proto	0.08	1	8	0.500	1.000	0.250	0.250
calib-proto	0.08	2	8	0.625	1.000	0.000	0.125
calib-proto	0.08	3	8	0.500	1.000	0.000	0.375
calib-proto	0.08	4	8	0.625	1.000	0.125	0.125
calib-proto	0.08	5	8	0.875	0.875	0.000	0.125
calib-proto	0.08	6	8	0.375	1.000	0.125	0.375
calib-proto	0.08	7	8	0.500	0.875	0.375	0.250
LFI-v5	0.08	0	8	0.125	0.625	0.500	0.500
LFI-v5	0.08	1	8	0.250	0.750	0.250	0.625
LFI-v5	0.08	2	8	0.250	0.750	0.375	0.625
LFI-v5	0.08	3	8	0.250	0.625	0.375	0.625
LFI-v5	0.08	4	8	0.500	0.750	0.375	0.250
LFI-v5	0.08	5	8	0.250	0.750	0.625	0.375
LFI-v5	0.08	6	8	0.125	0.625	0.125	0.875
LFI-v5	0.08	7	8	0.375	0.875	0.250	0.250
particle	0.08	0	8	0.375	0.625	0.250	0.375
particle	0.08	1	8	0.750	1.000	0.000	0.250
particle	0.08	2	8	0.250	1.000	0.250	0.375
particle	0.08	3	8	0.500	0.875	0.250	0.250
particle	0.08	4	8	0.625	0.875	0.000	0.250
particle	0.08	5	8	0.750	0.875	0.125	0.250

Table: Seed-level fixed-risk metrics continued

Method	Budget	Seed	Episodes	Success	Fixture	Damage	Repeat
particle	0.08	6	8	0.375	0.750	0.250	0.500
particle	0.08	7	8	0.500	0.875	0.250	0.500
proto-ID	0.08	0	8	0.375	0.875	0.250	0.250
proto-ID	0.08	1	8	0.500	1.000	0.000	0.250
proto-ID	0.08	2	8	0.750	1.000	0.125	0.125
proto-ID	0.08	3	8	0.625	1.000	0.125	0.250
proto-ID	0.08	4	8	0.750	1.000	0.125	0.125
proto-ID	0.08	5	8	0.500	0.875	0.375	0.250
proto-ID	0.08	6	8	0.500	1.000	0.250	0.125
proto-ID	0.08	7	8	0.500	0.875	0.125	0.375
rf-probe	0.08	0	8	0.375	0.750	0.375	0.625
rf-probe	0.08	1	8	0.750	1.000	0.125	0.000
rf-probe	0.08	2	8	0.750	1.000	0.125	0.125
rf-probe	0.08	3	8	0.500	1.000	0.250	0.125
rf-probe	0.08	4	8	0.500	0.875	0.250	0.500
rf-probe	0.08	5	8	0.625	0.875	0.125	0.375
rf-probe	0.08	6	8	0.500	1.000	0.125	0.375
rf-probe	0.08	7	8	0.625	1.000	0.125	0.125
robust-imp	0.08	0	8	0.125	0.875	0.625	0.625
robust-imp	0.08	1	8	0.625	1.000	0.125	0.250
robust-imp	0.08	2	8	0.500	1.000	0.250	0.500
robust-imp	0.08	3	8	0.250	1.000	0.375	0.500
robust-imp	0.08	4	8	0.500	1.000	0.250	0.250
robust-imp	0.08	5	8	0.500	0.875	0.250	0.500
robust-imp	0.08	6	8	0.375	0.875	0.375	0.500
robust-imp	0.08	7	8	0.625	0.875	0.125	0.375
adaptive-probe	0.12	0	8	0.500	0.875	0.125	0.500
adaptive-probe	0.12	1	8	1.000	1.000	0.000	0.000
adaptive-probe	0.12	2	8	0.875	1.000	0.000	0.125
adaptive-probe	0.12	3	8	0.500	1.000	0.125	0.500
adaptive-probe	0.12	4	8	0.375	0.875	0.125	0.625
adaptive-probe	0.12	5	8	0.875	1.000	0.000	0.125
adaptive-probe	0.12	6	8	0.875	0.875	0.125	0.125
adaptive-probe	0.12	7	8	0.750	1.000	0.000	0.250
bayes	0.12	0	8	0.750	0.875	0.000	0.125
bayes	0.12	1	8	0.500	0.875	0.125	0.375
bayes	0.12	2	8	0.250	0.875	0.250	0.625
bayes	0.12	3	8	0.250	0.875	0.250	0.625

Table: Seed-level fixed-risk metrics continued

Method	Budget	Seed	Episodes	Success	Fixture	Damage	Repeat
bayes	0.12	4	8	0.500	0.750	0.250	0.375
bayes	0.12	5	8	0.375	1.000	0.125	0.500
bayes	0.12	6	8	0.375	0.875	0.375	0.500
bayes	0.12	7	8	0.500	0.875	0.250	0.375
calib-proto	0.12	0	8	0.375	0.875	0.250	0.375
calib-proto	0.12	1	8	0.625	1.000	0.125	0.250
calib-proto	0.12	2	8	0.500	1.000	0.000	0.250
calib-proto	0.12	3	8	0.625	1.000	0.000	0.250
calib-proto	0.12	4	8	0.750	0.875	0.125	0.125
calib-proto	0.12	5	8	0.750	1.000	0.000	0.250
calib-proto	0.12	6	8	0.500	1.000	0.125	0.250
calib-proto	0.12	7	8	0.750	0.875	0.125	0.125
LFI-v5	0.12	0	8	0.125	0.750	0.500	0.625
LFI-v5	0.12	1	8	0.250	0.750	0.250	0.500
LFI-v5	0.12	2	8	0.375	0.625	0.375	0.500
LFI-v5	0.12	3	8	0.250	0.750	0.125	0.625
LFI-v5	0.12	4	8	0.375	0.750	0.250	0.500
LFI-v5	0.12	5	8	0.375	0.750	0.250	0.375
LFI-v5	0.12	6	8	0.125	0.750	0.125	0.875
LFI-v5	0.12	7	8	0.250	0.750	0.375	0.375
particle	0.12	0	8	0.375	0.750	0.375	0.375
particle	0.12	1	8	0.625	0.875	0.125	0.375
particle	0.12	2	8	0.625	1.000	0.125	0.250
particle	0.12	3	8	0.500	0.875	0.375	0.125
particle	0.12	4	8	0.625	0.750	0.125	0.375
particle	0.12	5	8	0.750	1.000	0.125	0.250
particle	0.12	6	8	0.625	0.875	0.250	0.375
particle	0.12	7	8	0.625	0.875	0.125	0.375
proto-ID	0.12	0	8	0.250	0.875	0.125	0.500
proto-ID	0.12	1	8	0.500	1.000	0.250	0.125
proto-ID	0.12	2	8	0.750	1.000	0.250	0.000
proto-ID	0.12	3	8	0.625	1.000	0.125	0.250
proto-ID	0.12	4	8	0.500	0.875	0.125	0.500
proto-ID	0.12	5	8	0.500	1.000	0.250	0.250
proto-ID	0.12	6	8	0.625	1.000	0.125	0.125
proto-ID	0.12	7	8	0.250	0.875	0.125	0.750
rf-probe	0.12	0	8	0.250	0.750	0.125	0.625
rf-probe	0.12	1	8	0.875	1.000	0.000	0.000

Table: Seed-level fixed-risk metrics continued

Method	Budget	Seed	Episodes	Success	Fixture	Damage	Repeat
rf-probe	0.12	2	8	0.250	1.000	0.250	0.625
rf-probe	0.12	3	8	0.625	1.000	0.250	0.125
rf-probe	0.12	4	8	0.375	0.875	0.250	0.625
rf-probe	0.12	5	8	0.625	1.000	0.000	0.375
rf-probe	0.12	6	8	0.750	1.000	0.000	0.250
rf-probe	0.12	7	8	0.625	1.000	0.125	0.125
robust-imp	0.12	0	8	0.375	0.875	0.375	0.375
robust-imp	0.12	1	8	0.500	1.000	0.125	0.500
robust-imp	0.12	2	8	0.375	1.000	0.125	0.500
robust-imp	0.12	3	8	0.250	1.000	0.625	0.500
robust-imp	0.12	4	8	0.250	0.875	0.250	0.375
robust-imp	0.12	5	8	0.500	1.000	0.125	0.375
robust-imp	0.12	6	8	0.625	0.875	0.250	0.250
robust-imp	0.12	7	8	0.500	0.875	0.125	0.500
adaptive-probe	0.18	0	8	0.375	0.875	0.000	0.625
adaptive-probe	0.18	1	8	0.875	1.000	0.000	0.125
adaptive-probe	0.18	2	8	0.375	1.000	0.375	0.500
adaptive-probe	0.18	3	8	0.375	0.875	0.250	0.500
adaptive-probe	0.18	4	8	0.750	0.875	0.125	0.125
adaptive-probe	0.18	5	8	0.750	1.000	0.125	0.250
adaptive-probe	0.18	6	8	0.875	0.875	0.125	0.125
adaptive-probe	0.18	7	8	0.625	0.875	0.000	0.375
bayes	0.18	0	8	0.625	0.875	0.000	0.375
bayes	0.18	1	8	0.375	0.750	0.125	0.500
bayes	0.18	2	8	0.375	1.000	0.125	0.625
bayes	0.18	3	8	0.500	0.750	0.375	0.375
bayes	0.18	4	8	0.375	0.750	0.500	0.375
bayes	0.18	5	8	0.500	0.875	0.125	0.375
bayes	0.18	6	8	0.750	0.750	0.250	0.250
bayes	0.18	7	8	0.375	0.875	0.125	0.500
calib-proto	0.18	0	8	0.250	0.875	0.125	0.500
calib-proto	0.18	1	8	0.875	1.000	0.000	0.125
calib-proto	0.18	2	8	0.750	1.000	0.125	0.125
calib-proto	0.18	3	8	0.750	0.875	0.000	0.125
calib-proto	0.18	4	8	0.750	0.875	0.000	0.250
calib-proto	0.18	5	8	0.625	1.000	0.000	0.375
calib-proto	0.18	6	8	0.500	0.875	0.000	0.500
calib-proto	0.18	7	8	0.625	0.875	0.125	0.250

Table: Seed-level fixed-risk metrics continued

Method	Budget	Seed	Episodes	Success	Fixture	Damage	Repeat
LFI-v5	0.18	0	8	0.250	0.875	0.750	0.500
LFI-v5	0.18	1	8	0.375	0.625	0.125	0.375
LFI-v5	0.18	2	8	0.250	0.500	0.125	0.750
LFI-v5	0.18	3	8	0.125	0.625	0.375	0.750
LFI-v5	0.18	4	8	0.375	0.750	0.250	0.375
LFI-v5	0.18	5	8	0.250	0.875	0.250	0.500
LFI-v5	0.18	6	8	0.375	0.750	0.375	0.375
LFI-v5	0.18	7	8	0.125	0.625	0.500	0.750
particle	0.18	0	8	0.625	0.875	0.375	0.125
particle	0.18	1	8	0.375	0.875	0.375	0.250
particle	0.18	2	8	0.625	0.875	0.250	0.125
particle	0.18	3	8	0.500	0.875	0.250	0.250
particle	0.18	4	8	0.500	0.750	0.250	0.500
particle	0.18	5	8	0.625	1.000	0.000	0.250
particle	0.18	6	8	0.625	0.875	0.125	0.250
particle	0.18	7	8	0.375	0.875	0.250	0.375
proto-ID	0.18	0	8	0.875	0.875	0.000	0.125
proto-ID	0.18	1	8	0.500	1.000	0.000	0.375
proto-ID	0.18	2	8	0.500	0.875	0.125	0.375
proto-ID	0.18	3	8	0.375	0.875	0.250	0.375
proto-ID	0.18	4	8	0.500	0.875	0.125	0.500
proto-ID	0.18	5	8	0.250	1.000	0.125	0.500
proto-ID	0.18	6	8	0.625	0.875	0.375	0.125
proto-ID	0.18	7	8	0.375	0.875	0.125	0.625
rf-probe	0.18	0	8	0.375	0.875	0.000	0.375
rf-probe	0.18	1	8	0.750	1.000	0.125	0.250
rf-probe	0.18	2	8	0.250	0.875	0.375	0.500
rf-probe	0.18	3	8	0.750	0.875	0.000	0.125
rf-probe	0.18	4	8	0.500	0.875	0.125	0.375
rf-probe	0.18	5	8	0.375	1.000	0.125	0.500
rf-probe	0.18	6	8	0.500	0.750	0.250	0.375
rf-probe	0.18	7	8	0.500	0.875	0.125	0.500
robust-imp	0.18	0	8	0.500	1.000	0.375	0.250
robust-imp	0.18	1	8	0.500	1.000	0.250	0.375
robust-imp	0.18	2	8	0.250	1.000	0.250	0.500
robust-imp	0.18	3	8	0.375	0.875	0.375	0.375
robust-imp	0.18	4	8	0.250	0.875	0.125	0.625
robust-imp	0.18	5	8	0.625	1.000	0.000	0.250

Table: Seed-level fixed-risk metrics continued

Method	Budget	Seed	Episodes	Success	Fixture	Damage	Repeat
robust-imp	0.18	6	8	0.500	0.875	0.250	0.500
robust-imp	0.18	7	8	0.625	0.875	0.250	0.375
adaptive-probe	0.25	0	8	0.625	1.000	0.000	0.375
adaptive-probe	0.25	1	8	0.625	0.875	0.125	0.125
adaptive-probe	0.25	2	8	0.625	1.000	0.000	0.375
adaptive-probe	0.25	3	8	0.625	0.875	0.125	0.250
adaptive-probe	0.25	4	8	0.625	1.000	0.000	0.250
adaptive-probe	0.25	5	8	0.875	1.000	0.125	0.125
adaptive-probe	0.25	6	8	0.750	1.000	0.125	0.125
adaptive-probe	0.25	7	8	0.625	1.000	0.000	0.375
bayes	0.25	0	8	0.375	0.875	0.125	0.625
bayes	0.25	1	8	0.500	0.750	0.250	0.375
bayes	0.25	2	8	0.750	1.000	0.000	0.125
bayes	0.25	3	8	0.500	0.750	0.250	0.375
bayes	0.25	4	8	0.500	0.875	0.125	0.375
bayes	0.25	5	8	0.625	0.875	0.125	0.375
bayes	0.25	6	8	0.750	0.875	0.125	0.250
bayes	0.25	7	8	0.375	0.875	0.125	0.500
calib-proto	0.25	0	8	0.375	1.000	0.125	0.500
calib-proto	0.25	1	8	0.250	0.875	0.500	0.500
calib-proto	0.25	2	8	0.625	1.000	0.250	0.125
calib-proto	0.25	3	8	0.750	0.875	0.000	0.125
calib-proto	0.25	4	8	0.375	1.000	0.000	0.375
calib-proto	0.25	5	8	0.625	1.000	0.250	0.125
calib-proto	0.25	6	8	0.750	1.000	0.125	0.125
calib-proto	0.25	7	8	0.750	1.000	0.125	0.000
LFI-v5	0.25	0	8	0.375	0.625	0.375	0.500
LFI-v5	0.25	1	8	0.125	0.500	0.750	0.625
LFI-v5	0.25	2	8	0.125	0.875	0.125	0.500
LFI-v5	0.25	3	8	0.125	0.625	0.375	0.750
LFI-v5	0.25	4	8	0.125	0.625	0.625	0.500
LFI-v5	0.25	5	8	0.250	0.875	0.250	0.625
LFI-v5	0.25	6	8	0.125	0.750	0.625	0.375
LFI-v5	0.25	7	8	0.125	0.750	0.250	0.750
particle	0.25	0	8	0.625	0.875	0.250	0.250
particle	0.25	1	8	0.375	0.625	0.500	0.500
particle	0.25	2	8	0.375	1.000	0.125	0.375
particle	0.25	3	8	0.500	0.750	0.125	0.500

Table: Seed-level fixed-risk metrics continued

Method	Budget	Seed	Episodes	Success	Fixture	Damage	Repeat
particle	0.25	4	8	0.750	1.000	0.000	0.250
particle	0.25	5	8	0.375	0.875	0.000	0.500
particle	0.25	6	8	0.500	0.875	0.125	0.500
particle	0.25	7	8	0.375	0.875	0.250	0.500
proto-ID	0.25	0	8	0.500	1.000	0.000	0.250
proto-ID	0.25	1	8	0.250	0.875	0.250	0.750
proto-ID	0.25	2	8	0.500	1.000	0.000	0.375
proto-ID	0.25	3	8	0.625	0.875	0.125	0.250
proto-ID	0.25	4	8	0.875	1.000	0.125	0.000
proto-ID	0.25	5	8	0.750	1.000	0.000	0.250
proto-ID	0.25	6	8	0.750	1.000	0.000	0.250
proto-ID	0.25	7	8	0.500	1.000	0.000	0.500
rf-probe	0.25	0	8	0.625	1.000	0.000	0.375
rf-probe	0.25	1	8	0.125	0.625	0.500	0.750
rf-probe	0.25	2	8	0.625	1.000	0.250	0.000
rf-probe	0.25	3	8	0.625	0.875	0.125	0.250
rf-probe	0.25	4	8	0.375	0.875	0.000	0.500
rf-probe	0.25	5	8	0.500	0.875	0.125	0.375
rf-probe	0.25	6	8	0.625	1.000	0.000	0.250
rf-probe	0.25	7	8	0.625	1.000	0.000	0.375
robust-imp	0.25	0	8	0.750	1.000	0.125	0.250
robust-imp	0.25	1	8	0.125	0.875	0.125	0.750
robust-imp	0.25	2	8	0.500	1.000	0.375	0.375
robust-imp	0.25	3	8	0.625	0.875	0.000	0.125
robust-imp	0.25	4	8	0.500	1.000	0.125	0.375
robust-imp	0.25	5	8	0.250	1.000	0.625	0.375
robust-imp	0.25	6	8	0.625	1.000	0.375	0.125
robust-imp	0.25	7	8	0.500	1.000	0.250	0.375

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